

Preliminary validation of the 1D modeling of the DYNASTY natural circulation loop against results from water experimental campaign

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ABSTRACT

In the continuous improvements of the safety of nuclear power plants, the study of natural circulation is of primary importance for developing passive safety systems and Generation-IV reactors, such as the Molten Salt Fast Reactor (MSFR). In particular, removing the decay heat during the shutdown of the MSFR involves dealing with a circulating fuel driven by natural circulation; as such, the development and validation of natural circulation models are of primary importance. To validate models and provide experimental data on natural circulation with a distributed heat source, the DYNASTY (DYNamics of NATural circulation for molten Salt internally heated) natural circulation loop has been built on the premises of Politecnico di Milano. This paper focuses on improving the 1D modeling of natural circulation phenomena in the DYNASTY facility and validating them against the results (also reported in this work) found in the first experimental campaign, carried out with water as the working fluid. The modeling of DYNASTY is based on a 1D approach using DYMOLA®, a modeling environment based on the MODELICA simulation language. The models used are improved versions of the ones available in the literature, which better model the experimental facility as they now include the mass flow rate meter, the heat loss between the facility and the environment, and a more realistic cooler. The improved model was then used to study the outcomes of the model simulations and verify the influence of the different numerical integration algorithms on the simulation results. The first DYNASTY experimental campaign provides in-depth testing of its behavior, with different cooling fan speeds and considering all possible heating configurations. The collected results for the heating and cooling transients are the mass flow rate and four fluid temperatures measured by thermocouples placed at the beginning and the end of the four DYNASTY legs. The experimental results were compared with the model simulations, showing the good predicting capabilities of the latter one following the tuning of the heat exchange model. In particular, the comparison showed an excellent match between the model and the experimental results for the heating transient while showing improving margins for the cooling transient.

1. Introduction

Natural circulation is the physical phenomenon that describes the spontaneous flow in a fluid induced by buoyancy forces, caused by density variations because of temperature or phase changes (Bergman et al., 2011). Natural circulation is interesting for the nuclear industry since it can provide a passive way to remove the decay heat from the core without needing active systems. For this reason, it is currently used in the AP-1000 (Sutharshan et al., 2011) and in the ESBWR (Rassame et al., 2017) Generation-III+ nuclear reactors as a runaway system to dissipate the decay heat during the shutdown. The study of natural circulation is also significant when developing the safety systems for Generation IV nuclear reactors, such as the Molten Salt Fast Reactors (MSFR) (Serp et al., 2014). The adoption of natural circulation as a passive heat removal mechanism relies on its ability to maintain high

enough temperatures to achieve the required flow rate for efficient heat removal. Since natural circulation results from a balance between friction losses and buoyancy forces, a potential outcome could be an oscillating flow regime or even an inversion in the fluid's motion (Misale, 2014) which are conditions to avoid when designing a safety system.

The problem of natural circulation stability is a well-known issue (Welander, 1967; Lorenzi and Cammi, 2019; Schmid and Brandt, 2014) that arises from the competition between the buoyancy forces driving the fluid and the friction forces, which hinders the fluid motion, as shown in Fig. 1. These two effects heavily rely on the fluid thermal properties that depend on its temperature. The higher the temperature difference, the higher the mass flow rate induced by natural circulation. As the mass flow rate increases, the pressure losses and the heat transfer

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Nomenclature

Acronyms

BDF	Backward Differentiation Formula
CPU	Central Processing Unit
DAE	Differential Algebraic Equations
DH	Distributed Heat
DYNASTY	DYnamics of NATural circulation for molten SaLT internally heated
EHS	External Heat Source
ESDIRK	Explicit Singly Diagonal Implicit Runge–Kutta
FOM	Figure Of Merit
HHHC	Horizontal Heater Horizontal Cooler
IDE	Integrated Development Environment
IHG	Internal Heat Generation
MFRM	Mass Flow Rate Meter
MSFR	Molten Salt Fast Reactor
NCL	Natural Circulation Loop
O-O	Object Oriented
ODE	Ordinary Differential Equations
RK	Runge–Kutta
SDIRK	Singly Diagonal Implicit Runge–Kutta
VHHC	Vertical Heater Horizontal Cooler

Latin

$\dot{m}$	Mass flow rate (g s^{-1})
A	Surface area (m^2)
C_2	Fin efficiency calculation constant (–)
d, D	Diameter (m)
h	Convective heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
I_0, I_1	Second kind modified Bessel function (–)
k	Thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$)
K, K_1, K_2	Mass Flow Rate Meter localized pressure drop coefficients
K_0, K_1	First kind modified Bessel function (–)
m	Fin heat transfer coefficient (–)
Nu	Nusselt number (–)
p	Pressure (Pa)
Pr	Prandtl number (–)
Q	Heat flow rate (W)
r	radius (m)
Re	Reynolds number (–)
T	Temperature (K)

Greek

β	Thermal expansion coefficient (K^{-1})
Δ	Difference
ϵ	Emissivity coefficient (–)
η	Fin efficiency (–)
ν	Kinematic viscosity ($\text{m}^2 \text{s}^{-1}$)
Φ	Heat flux (W m^{-2})
ρ	Density (kg m^{-3})
σ_{SB}	Stephen Boltzmann constant ($\text{W m}^{-2} \text{K}^{-4}$)

Subscripts

<i>air</i>	Quantity referred to air
<i>D</i>	Diameter characteristic length
<i>fin</i>	Quantity referred to fins
<i>i</i>	<i>i</i> th order
<i>ii</i>	<i>ii</i> th node
<i>ins</i>	Quantity referred to DYNASTY insulator
<i>pipe</i>	Quantity referred to DYNASTY pipe
<i>t</i>	Total
<i>w</i>	Wall
<i>b</i>	Quantity referred to the bare pipe

Superscripts

–	Average value
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coefficient also increase. In particular, the growth of the heat transfer coefficient causes a further rise in the temperature differences in the fluid, which leads to a further increase in the mass flow rate, thus defining positive feedback. In contrast, the increase in pressure losses leads to a decrease in the mass flow rate, causing negative feedback.

In the MSFR, the presence of the fuel in a homogeneous mixture with the thermal carrier entails a very innovative topic: natural circulation with Internal Heat Generation (IHG) (Jeong et al., 2018). The issue of having natural circulation with IHG arises because the fissile fuel is in the liquid phase and is homogeneously mixed with the thermal carrier. Therefore, when designing MSFR reactors, a careful analysis of IHG, from the thermo-physical properties to the effect on mass flow and pressure losses, must be performed. Unfortunately, setting up an experimental facility with an IHG configuration is quite hard, as it would require either chemical or nuclear reactions. However, a facility

with an axial section long enough to consider the radial one negligible represents a reasonable first approximation, to neglect the difference between IHG and an External Heat Source (EHS) (Cammi et al., 2016b).

Natural circulation Loops (NCLs) are powerful tools for studying natural circulation in the presence of IHG. Usually, they present the axial pipe lengths much larger than the radial ones so that, as a first approximation, the variation of physical quantities in the radial direction can be neglected (Cammi et al., 2016b). This approximation allows setting up an experimental facility using an EHS rather than IHG while still being able to compare the results found with numerical models using both EHS and IHG. NCLs are also very useful for studying the stability and behavior of natural circulation. In particular, in single-phase NCLs, such as the Bhaba Single Phase Natural Circulation Loop (Vijayan et al., 2001; Vijayan, 2002) and the L2 loop (Misale and Garibaldi, 2010), the flow behavior is mainly affected by density wave instabilities (Lorenzi and Cammi, 2019). The characteristic velocity of such phenomena strongly depends on local properties as temperature and pressure losses. In 1967, Welander (1967) investigated the problem of natural circulation loop instability by proposing a model based on the growth of small oscillation because of the formation of hot and cold wave packets inside the system. The amplification of hot wave packets that pass through a heat source leads to a positive contribution to the buoyancy flow that makes the mass flow rate reach its maximum. Instead, when the hot wave packets pass through the heat sink, the buoyancy flow receives a negative contribution that causes the mass flow rate to decrease to its minimum value. This decrease then enables the formation of a new hot packet in the heat source. The same reasoning applies to the cold wave packet, and the fluid behaves like a pendulum. Fig. 2 shows the mechanism of the wave-packet instability, also called “Welander instabilities”.

DYNASTY (DYnamics of NATural circulation for molten SaLT internally heated) is a large NCL purposely built to study the influence of a distributed heat source on the behavior of natural circulation (Cammi et al., 2016a; Pini, 2017; Cauzzi, 2019). DYNASTY is a flexible facility that permits using either a Distributed Heat (DH) source or a Localized

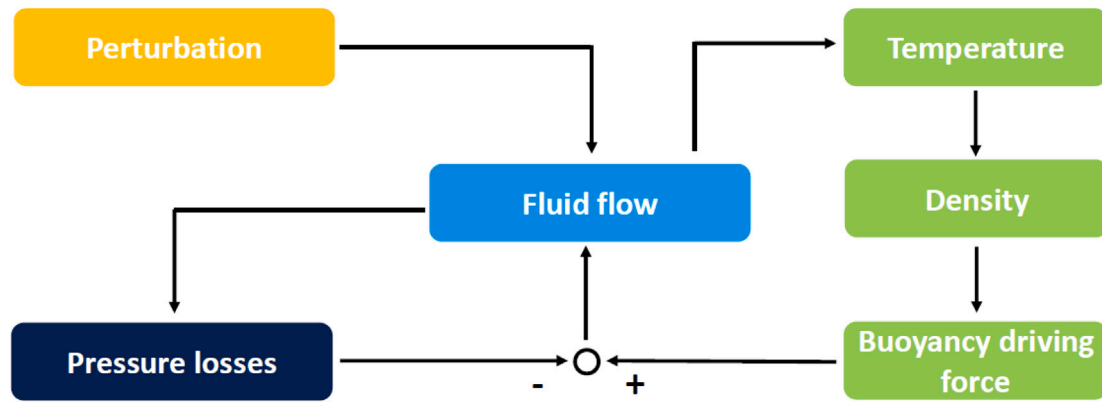


Fig. 1. Natural Circulation feedback scheme (Lorenzi and Cammi, 2019).



Fig. 2. Welander instability mechanism, red and blue zones respectively represents hot and cold wave-packets (Welander, 1967).

Heat (LH) source, according to the experiment. In particular, the different LH configurations that DYNASTY can adopt during experiments are Vertical Heater Horizontal Cooler (VHHC) and Horizontal Heater Horizontal Cooler (HHHC) sources. The facility is realized in AISI316 stainless steel to sustain the high temperature expected when operating with molten salts (Pini, 2017). Although DYNASTY was designed mainly to work with molten salts, it can also use different fluids, such as glycol and water. Fig. 3 shows a picture of the DYNASTY facility.

DYNASTY modeling has been previously tackled both with a 1D approach (Cammi et al., 2016a; Cauzzi, 2019) and with a 3D one (Cauzzi, 2019; Battistini, 2020; Battistini et al., 2021). This paper focuses on the 1D modeling of the system, adopting the DYMOLA development environment based on the Modelica language (Dassault Systèmes, 2020). Modelica is an Object-Oriented (O-O) simulation language that uses physical and engineering principles, like energy and mass balance equations, to model complex systems (Fritzson, 2014; Tiller, 2019). This work aims at testing and validating the 1D DYMOLA® models of the DYNASTY facility. The models used were developed in previous works (Cauzzi, 2019; Cammi et al., 2016b; Pini, 2017) and changed to include the mass flow rate meter pressure losses and the heat exchange between the facility and the environment. The testing activities on the model allowed the development of a preliminary experimental matrix to probe the behavior of the DYNASTY facility. This experimental campaign is the first in-depth analysis performed on the facility (Benzioni et al., 2022). DYNASTY aims at studying Natural Circulation in presence of a distributed heat source. In this condition, all the heaters are turned on to represent the distributed heat source, while the flow of the working fluid is driven by natural circulation. Therefore, the primary focus of this work is to compare the behavior of the facility and the models during the heating transient and to provide a preliminary validation of the developed 1D models. However, useful data can also be derived from the cooling transient regarding the behavior of a natural circulating fluid that was heated with a distributed heat source

when only the heat sink is active. So that in this paper the experiment and the model results are analyzed also in the cooling transient.

The paper structure is as follows. Section 2 introduces and analyzes the performances of the DYNASTY model used in this work. This model is based on the previously developed models available in literature (Cammi et al., 2016a; Pini, 2017; Cauzzi, 2019) and extended to better represent the actual facility. Section 3 describes the first DYNASTY experimental campaign, which spans all the different configurations of the facility and both the heating and cooling transient. Section 4 compares the experimental results with the DYMOLA models in both the heating and the cooling transients for the different configurations of the facility. Finally, Section 5 discusses the main findings of this work, along with possible further development.

2. DYNASTY modelica modeling

This section describes the 1D modeling approach to natural circulation in the DYNASTY facility. The modeling of the DYNASTY facility was carried out in DYMOLA, an Integrated Development Environment (IDE) based on the simulation language Modelica (Fritzson, 2014). Modelica is an object-oriented system code that provides a modeling alternative to more complex ones, especially when facing system-level simulations. It allows the description of single system components (or objects) directly in terms of physical equations and principles and connects them through standardized interfaces (or connectors). In addition, the a-causal component-based modeling strategy provides high re-usability of the models and flexibility of the MSFR plant configuration, as well as a more realistic description of the plant and offers the possibility to couple, in a straightforward way, the different physics presents (Fritzson, 2014). Thus, Modelica models are highly flexible, both in conditions and configurations, without the necessity of rebuilding the model from scratch. The drawback of the a-causal approach is that a Differential Algebraic Equation (DAE) system has to be solved. To do that, DYMOLA first translates the DAE system into a stiff Ordinary Differential Equation (ODE) system using a changed version of the Pantelides algorithm (Pantelides, 1988) and then solves the ODE system using the chosen numerical integration algorithm, e.g. DASSL (Petzold, 1982).

The DYNASTY model used in this work is shown in Fig. 4. This model accounts for the localized pressure drops because of the Mass Flow Rate Meter (MFRM) (Endress-Hausser, 2017), the thermal losses between the facility and the environment, and the increase in the cooler's surface because of the finning, the process used to develop the DYNASTY final model is described in Appendix. The model has been developed strictly for single-phase fluids so that it cannot model any phase change or two-phase flow. In the model, the components with the suffix "L" correspond to the DYNASTY pipes which can either provide the heat (the ones with the red arrows) or remove heat (LC1 and L4).



Fig. 3. DYNASTY natural circulation loop.

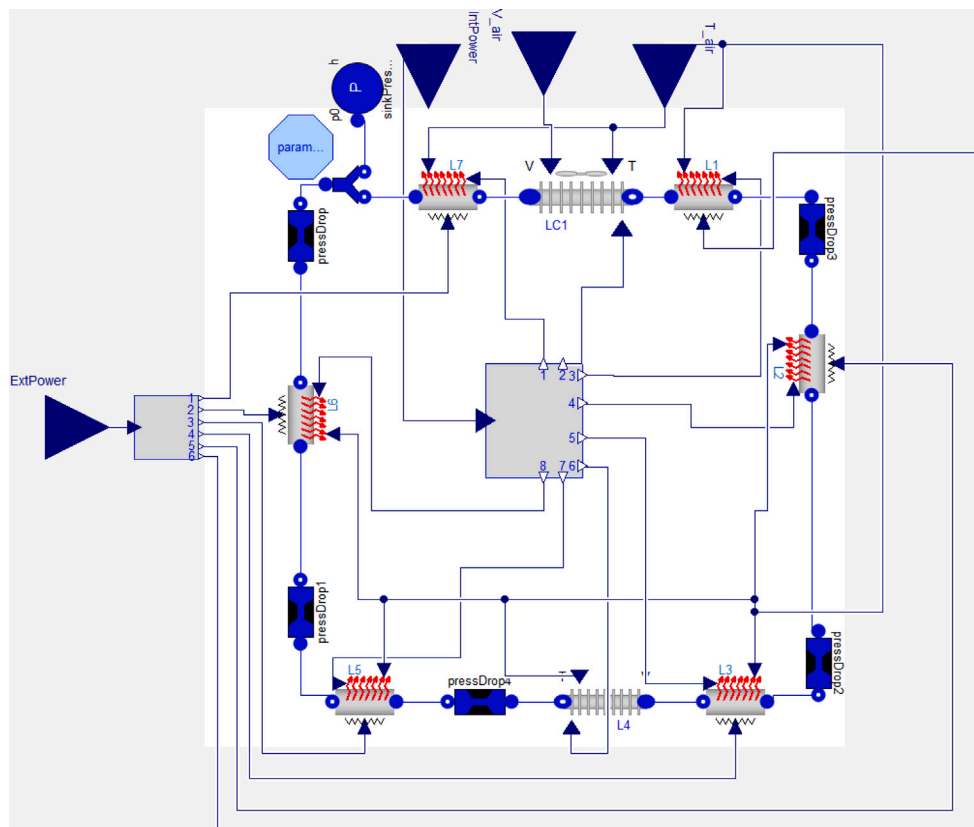


Fig. 4. DYNASTY final model.

This model accounts for the non-insulated and non-heated section of the MFRM and the two sections of the heated pipe before the cooler, $L1$ and $L7$. The component $L4$ represents the non-insulated and non-heated

section of the MFRM that uses Eq. (A.2) to model the heat losses. The model also has four external inputs for improved efficiency in setting up the simulations:

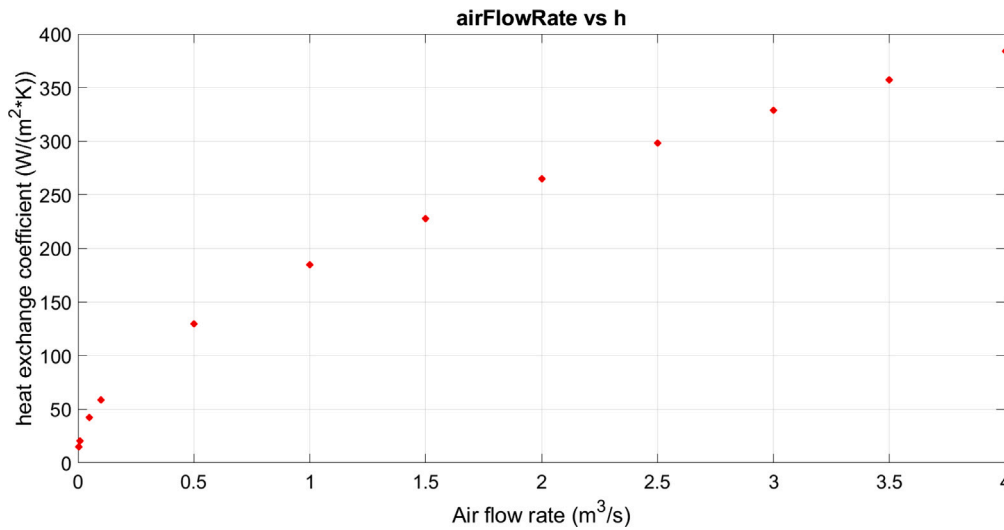


Fig. 5. Fan speed test results for the heat transfer coefficient h .

- *ExtPower* which provides the EH power signal;
- *IntPower* which provides the IHG power signal;
- *V_{air}* which provides the fan velocity signal to the cooler;
- *T_{air}* which provides the air temperature signal, taken as a fixed value that the model then uses to calculate a “film” temperature which is the average between the air temperature at the bulk (T_{air}) and at the pipe surface.

2.1. DYNASTY model performance analysis

The developed DYNASTY model behavior was studied and analyzed by setting up different simulations to test the impact of boundary conditions on the model performances. The conditions tested are the cooling fan speed, air temperature, and the pressure drop coefficient of the MFRM. These boundary conditions were chosen to provide useful data to set up experiments in the DYNASTY facility and test the influence on new components – and thus pressure losses – in the loop. The other boundary conditions used for the simulations are: the air temperature is set at 22 °C (except in the case with different air temperatures), the power delivered is 450 W with 5/8 of it in GV1 (*L7* and *L6*) and the remaining 3/8 in GV2 (*L1* and *L2*) while GO1, composed by *L3*, *L4* and *L5*, is not powered. Last, in this model *TC1* and *TC2* are fluid temperatures taken at the inlet and outlet section of the cooler *LC1*. The results reported in the figures correspond to the values found when the simulations reach a steady state under free dynamics.

The first simulations performed were the ones testing the DYNASTY behavior with different cooling fan velocities, or air flow rates, starting from a value of $0.005 \frac{m^3}{s}$ up to the maximum of $4 \frac{m^3}{s}$, with an increasing step size. Airflow rate values lower than $0.005 \frac{m^3}{s}$ cause the simulation to crash due to too high temperature reached in the simulations. This test was developed to understand where the fan is most effective in the DYNASTY operative range. Simulations results are collected in Figs. 5 and 6.

At low fan speed, the mass flow rate is more affected than at higher values, especially for from h_{cooler} values ranging from ~ 15 to $\sim 100 \frac{W}{m^2 \cdot K}$. The temperature values follow the same behavior passing from 90 °C to 50 °C in the first points, showing the largest change in the interval between 0 and $0.5 \frac{m^3}{s}$. Interestingly, the two temperatures decrease is not simultaneous. Fig. 5 gives a representation of the influence of the fan speed on the cooler heat transfer coefficient. As observed, the curve resembles a logarithmic in the first few points and then becomes almost linear following the Churchill–Bernstein correlation (Bergman et al., 2011) dependence on the air Reynolds number and thus on the fan speed. Also, as soon as the heat transfer coefficient increase becomes

almost linear, the mass flow rate of the fluid reaches a plateau where its value is almost independent of the value of the fan speed (as in the Churchill–Bernstein correlation Bergman et al., 2011).

In the next test, simulations analyzing the DYNASTY model behavior with different values of the MFRM pressure drop coefficient K , starting from 6 and ending at 350, were performed. In these simulations, the maximum fan air flow rate of $4 \frac{m^3}{s}$ is used. The results are collected in Fig. 7.

These simulation outcomes are expected. Increasing the pressure drops leads to a decrease in the mass flow rate and an increase in the temperature difference between *TC1* and *TC2*. Also, in this case, the highest change is seen with the first values of the MFRM K , having somewhat logarithmic behavior. The temperature difference curve has an almost linear increase with K because of a quasi-linear increase in the *TC1* values and a similar decrease in the *TC2* values. Thus, by increasing the overall pressure losses in the system, the temperature difference also increases. However, the increase in the temperature difference is not enough to compensate for the increment in the pressure drop coefficient, leading to the decreasing behavior in the mass flow rate.

Last, different air temperatures were tested, with the MFRM pressure drop coefficient set to its reference value and the fan speed set to maximum (air flow rate of $4 \frac{m^3}{s}$). The range of air temperature values considered goes from 10 °C up to 34 °C. The simulations results are collected in Fig. 8.

The results from these simulations show that the increase in the air temperature reflects an increase in cooler inlet and outlet temperatures, despite the decrease in the temperature difference. However, despite the temperature difference across the cooler decreases, the mass flow rate in the system increases. This increase compensates for the decrease in the temperature difference, allowing the system to exchange the same quantity of heat.

3. DYNASTY first experimental campaign

The first experimental campaign carried out on DYNASTY aims at fully characterizing the facility behavior with water as the working fluid (Benzoni et al., 2022). The experimental campaign spans all the possible heating configurations (two VHH, one HHH, and one DH) and with different values of the cooling fan speed (from 0% to 100% of the maximum fan speed, corresponding to an airflow rate of $4 \frac{m^3}{s}$, with a 25% step increase).

Table 1 shows the developed experimental matrix for this preliminary step of the experimental campaign, with the time of the heating

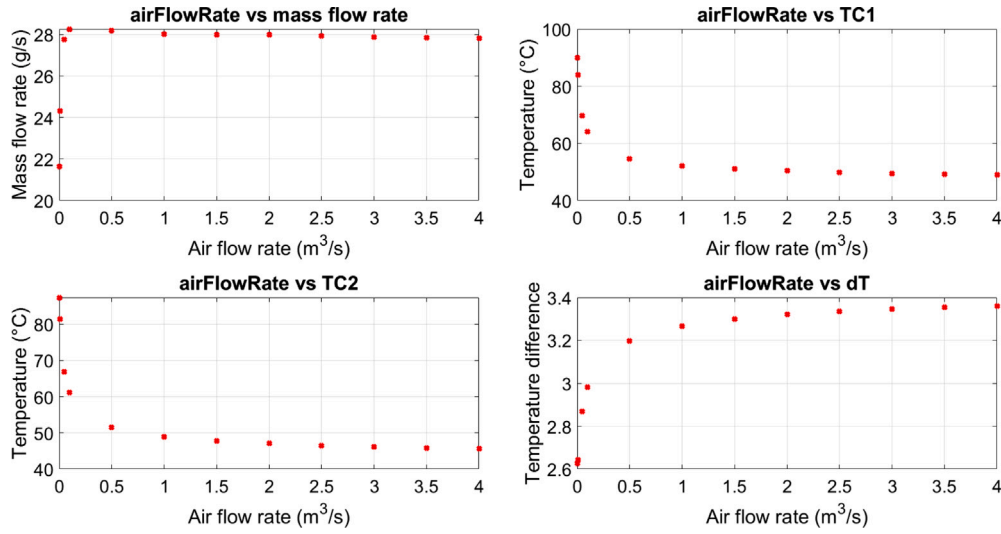


Fig. 6. Fan speed test results, mass flow rate, TC1, TC2 and temperature difference.

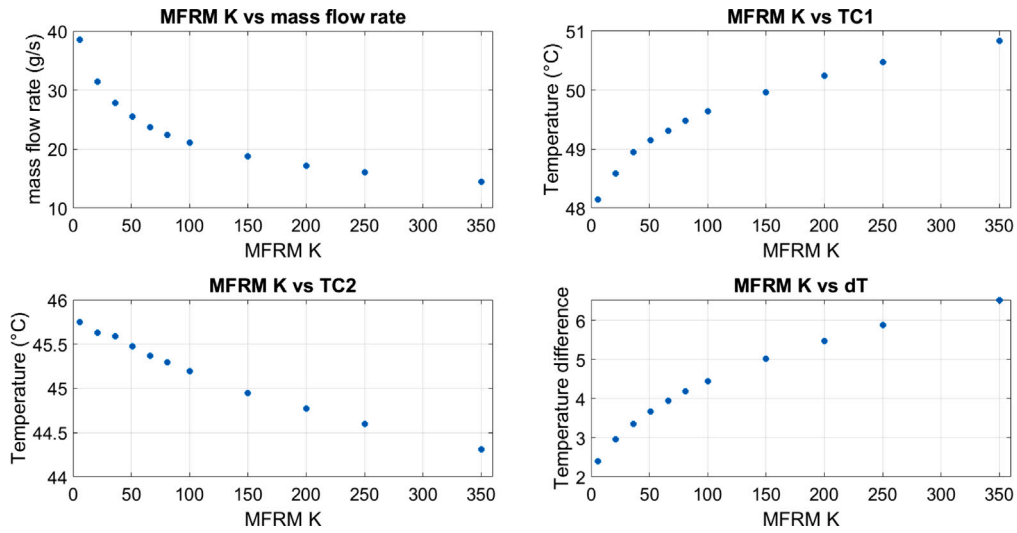


Fig. 7. MFRM pressure drop coefficient test results, mass flow rate, TC1, TC2 and temperature difference.

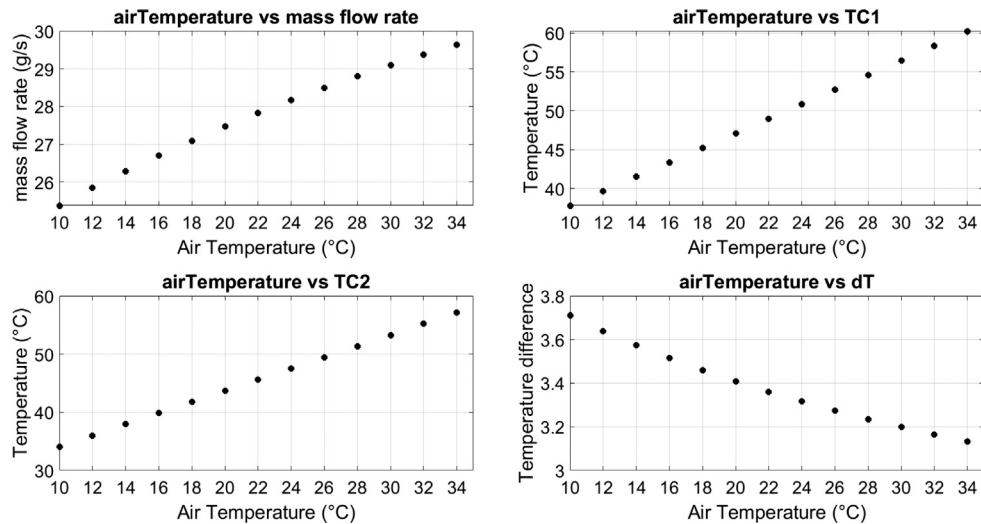


Fig. 8. Different air Temperature test results, mass flow rate, TC1, TC2 and temperature difference.

Table 1
DYNASTY experimental matrix.

Fan speed	GV1	Time (s)	GV2	Time (s)	GO1	Time (s)	DH	Time (s)
0%	✓	~15000	✓	~15100	✓	~15000	X	N.A
25%	✓	~15000	✓	~15000	✓	~15700	X	N.A
50%	✓	~15000	✓	~15000	✓	~15000	✓	~7000
75%	✓	~14500	✓	~15300	✓	~14500	✓	~9800
100%	X	N.A.	X	N.A.	X	N.A.	✓	~12000

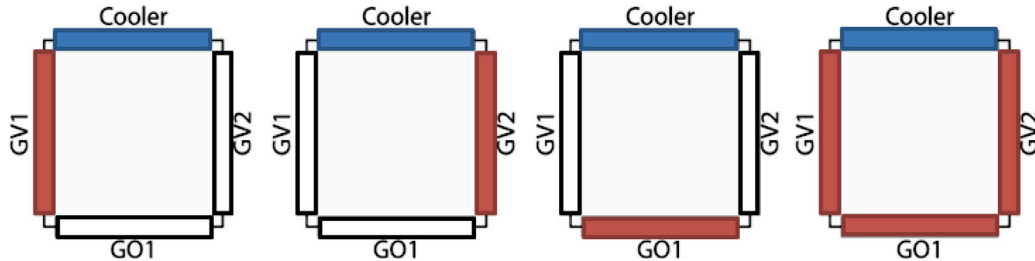


Fig. 9. DYNASTY heating configurations, red sections refer to heated sections.

transient for each experimental case. In the matrix, ✓ corresponds to a performed experiment, while X corresponds to not performed experiments as some cases were not considered. For the DH cases, fan speeds lower than 50% were not considered as the model predicted, before crashing, temperatures as high as 110 °C. In the cases with a localized heat source, values of fan speed with 100% were not performed since, as observed in the experiments, the cases with the fan turned on exhibit similar behavior and similar steady-state values of the mass flow rate regardless of fan speed.

Concerning the experimental matrix, the two VHHHC configurations tested are GV1, DYNASTY left vertical leg, and GV2, DYNASTY right vertical leg, whereas the HHHC configuration instead uses the horizontal lower leg GO1. Lastly, the DH configuration relies on heating all the legs at once. In Fig. 10, a schematic view of the DYNASTY experimental configuration is shown, with the heating legs, the fluid thermocouples (TC) position, the cooler, and the mass flow rate meter. The facility also has a second horizontal leg GO2, which houses the pump for future forced circulation experiments, and thus is bypassed in current experiments. The different heating configurations are summarized in Fig. 9. The measuring system of the facility is composed of four type J thermocouples (TC) ELSI® Q1-S80-J00-A0200-S00 (ELSI, 1998) that measure the temperature in the center of the fluid and by one Hendress+Hauser® Promass F80 DN25 Coriolis flow meter (Endress-Hauser, 2017). The tolerance of the measuring system is 1.5 °C for the thermocouples and 1.5% of the read value for the mass flow rate.

In this first experimental campaign, the results were collected for both the heating and cooling transients (where all the heating elements are off while the cooling fan is on, according to the experiment). The latter were collected since they could provide useful insight on the natural circulation behavior in the facility, although they are not of primary importance for DYNASTY and the present work. The experiments have been divided into two transients, the heating, from 0 s to approx. 15,000 s, and the cooling transient, from approx. 15,000 s when the heating is shut down to the end of the data acquisition. For the relevant cases, the fan is turned on both during heating and cooling transients.

In these experiments, the facility is tested without the insulator layer to characterize the thermal losses, with the heating power delivered in steps up to the maximum value (collected in Table 2 for the different cases). The power values were found by measuring the voltage and resistance of the heating strips with a multimeter. Table 3 instead collects the applied heating power net of the heat losses, calculated using the heat balance equation for the case at 0% fan speed. The results show that the horizontal leg GO1 is most afflicted by the absence

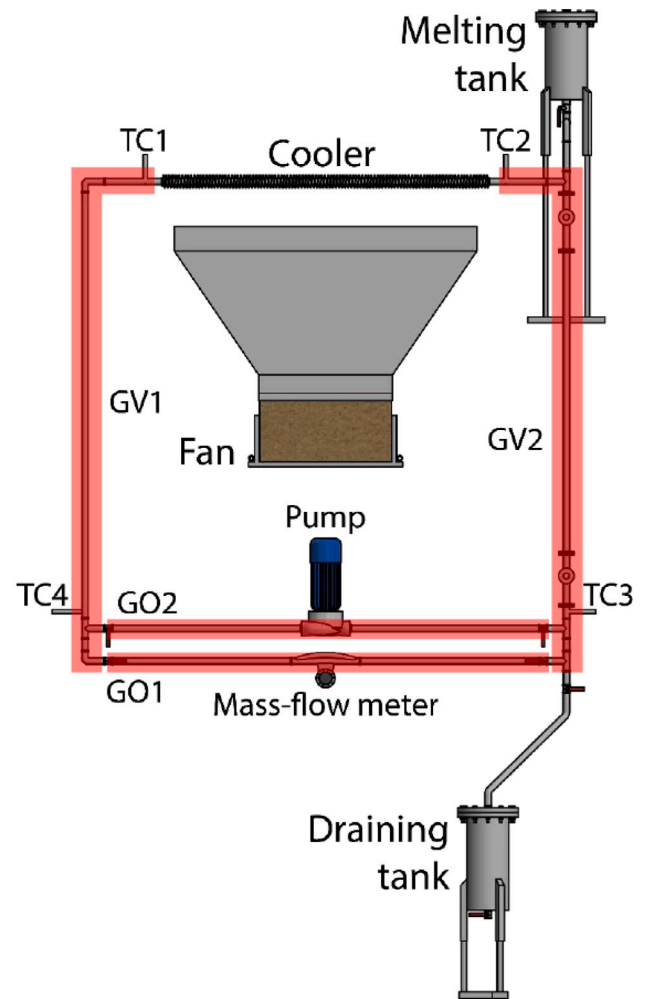


Fig. 10. DYNASTY experimental configuration, the heating sections are the one in red.

of the insulation layer, losing approximately 1.1 kW (almost 85% of the heating power) to the environment. The two vertical legs lose from 50% to 65% of the heating power. Last, the mass flow rate sign is assumed with a positive sign for clockwise flow.

Table 2
Measured applied power.

	GV1	GV2	GO1	DH (total)
Maximum heating power	2068 W	2071 W	1369 W	5508 W

Table 3
Net calculated power.

	GV1	GV2	GO1	DH (total)
Net heating power (at 0% fan speed)	1084 W	745 W	279 W	2108 W

3.1. DYNASTY preliminary experimental campaign results

3.1.1. Experimental results: VVHC case GV2

The first experimental results presented are the ones found in the VVHC configuration with GV2 turned on and are shown in Figs. 11 and 12, which collect respectively the evolution of the mass flow rate and the temperatures value read by the fluid thermocouples (TC) and by the mass flow rate meter (MFR) during the experiments. In the transient the heating power is provided as a step at its maximum value and is maintained until the mass flow rate curve reaches a stationary state, around 15,000 s. Then the power is turned off, again with a step. In the experimental results, The mass flow rate (Fig. 11) exhibits a negative sign, as the flow has a counter-clockwise direction (conventionally, the mass flow rate is positive if it flows in a clockwise direction). In all experiments, the natural circulation regime is established in the loop with different mass flow rates according to the adopted fan speed. The mass flow rate graphs show very little difference in the heating transients between the cases with the fan turned on at different speeds (from 25% to 75% of maximum fan speed), confirming the model's simulation results with different values of air flow rate (reported in Fig. 6). The mass flow rate changes drastically for the case with the cooling fan turned off in terms of the steady-state value found in the heating transient, whereas the curve shape is similar for all four cases. The drastic change in mass flow rate when the cooling fan is turned on is also observed in the DYMOLA[®] model performance test results with different air flow rates (Fig. 6). Both the model and the experiments show that the mass flow rate variations are higher at lower values of the airflow rate (and thus of the cooler heat transfer coefficient). This leads to the drastic increase in the stationary mass flow rate value observed when turning on the cooling fan compared to the case with the fan turned off (e.g. in the GV1 cases the mass flow rate goes from ~10 g/s to ~20 g/s when switching on the fan). In the experiments with the fan turned on, the mass flow rate increase is less prominent as the cooler convective heat transfer coefficient reaches the "threshold" value predicted by the model following the Churchill-Bernstein correlation (Bergman et al., 2011) (as shown in Fig. 6).

In the cooling transient, with the fan turned off, as soon as the power is switched off, the mass flow rate (Fig. 11) drastically decreases to almost zero, confirming the results of Cauzzi (2019). However, when the fan is turned on, the mass flow rate does not drop to zero after the initial oscillations, but shows a slow and smooth decay. The temperature graphs of Fig. 12 show the stratification of the temperature values in the facility, with TC2 being the hottest zone of the loop at the outlet of the heater section and TC3 the coldest zone at the inlet of the heater. The influence of the thermal losses of the non-insulated facility can be observed through the stratification of the temperature values along all the DYNASTY loops.

Fig. 13 shows a zoom for the mass flow rate and temperature curves in the cooling transient for the GV2 case with a fan speed of 50%. During the cooling transient, the fluid motion almost stops in the first minutes. This stop leads to an increase in the temperature difference of the cooler section (represented by TC1 and TC2) that reaches almost ten degrees, causing the mass flow rate curve to spike to a new maximum value, higher than the one seen in the heating transient. In the cooling

transient, this sudden increase in the value of the mass flow rate can be explained by the motion of a cold fluid packet (Welander, 1967) generated in the cooling section. After this peak, the mass flow rate curve shows dampened oscillations that lead to a decrease in its value that does not reach zero in the 25,000 s of the experimental transient.

3.1.2. Experimental results: VVHC case GV1

Next, the results for the VVHC configuration with GV1 turned on are shown, Figs. 14 and 15 which respectively show the mass flow rate and the fluid temperature evolution. The initial conditions are mostly the same as in the previous experiments, except for mild fluctuations in the environmental temperature that are reflected in the initial temperatures. Power is still provided as a step at its maximum value and is maintained until the mass flow rate curve reaches a stationary state, around 15,000 s. Then the power is turned off, again with a step.

The results of these heating transients are very similar to the ones for the GV2 case, with differences only in the final values and in the mass flow rate's sign (since DYNASTY is a symmetric facility) and in the maximum temperature values reached in the heating transient. The two VVHC configurations are almost the same, with only minor differences in the number of junctions (the GV2 leg has more of them, which leads to higher localized pressure losses). These higher losses explain the differences in the final value of the mass flow rate absolute value. The only major difference in the heating transient between the two cases is when the cooling fan is off (0% fan case). The reason for this difference is likely because the fluid temperature at the center of the pipe reaches 90 °C during the heating transient.

However, the inversion in the mass flow rate sign in the cooling transient was not expected. Fig. 16 shows the comparison of the two vertical cases in terms of mass flow rate for the case with a fan speed of 25% and further shows how the two vertical configurations behave symmetrically only during the heating transient, while in the cooling the two curves exhibit very similar behavior. Since both the VVHC cases show very similar behavior for the mass flow rate, it could mean that during the cooling transient, the mass flow rate is not affected by the heating configuration. A likely explanation for the preference in the flow direction in the cooling transient is the asymmetric distribution of localized pressure drops in the facility. The right vertical leg (GV2) houses more fittings than the left vertical one (GV1). So that during the initial part of the cooling, when the mass flow rate is at the minimum, the fluid moves towards the path with the least amount of pressure losses.

3.1.3. Experimental results: HHHC case GO1

The experimental results of the HHHC case (with GO1 turned on) are shown in Figs. 17 and 18. The experimental conditions are the same as the other two cases, with the only difference being that the HHHC configuration can deliver less power to the system compared to the VVHC configurations because the horizontal leg only has two heating strips instead of three. Another significant difference between the two configurations in DYNASTY, aside from the elevation of the outlet regarding the inlet, is the presence of significant pressure and thermal losses between the two heated horizontal sections, represented by the MFRM (which is not heated).

Despite having less power delivered to the system, the mass flow rate value reached is higher than in the VVHC cases. In the HHHC

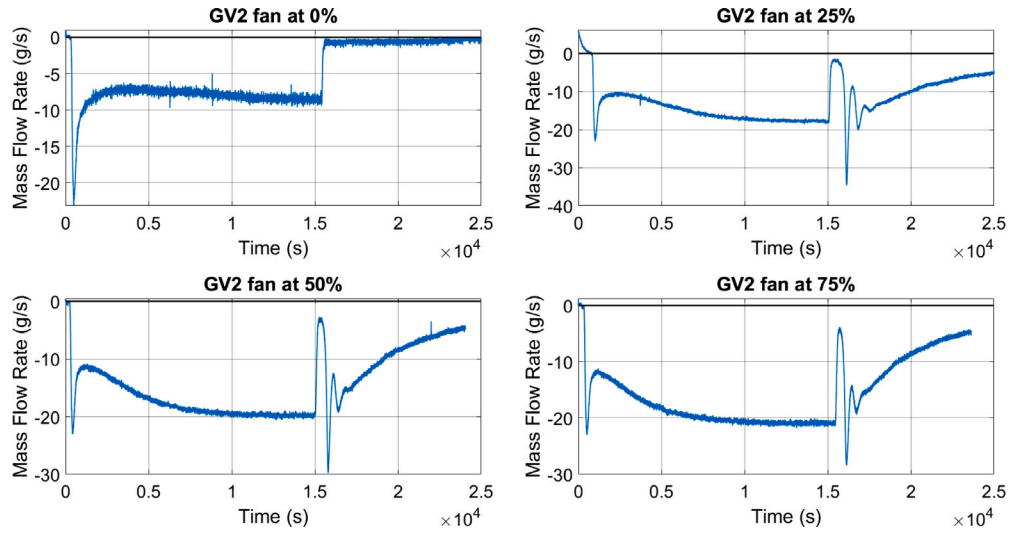


Fig. 11. GV2 results, mass flow rate evolution.

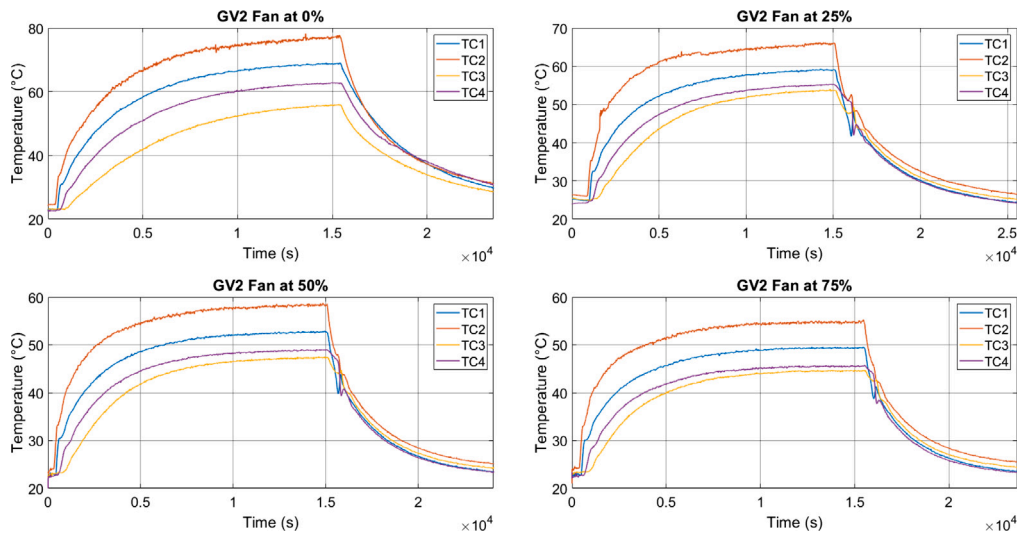


Fig. 12. GV2 results, fluid temperatures evolution.

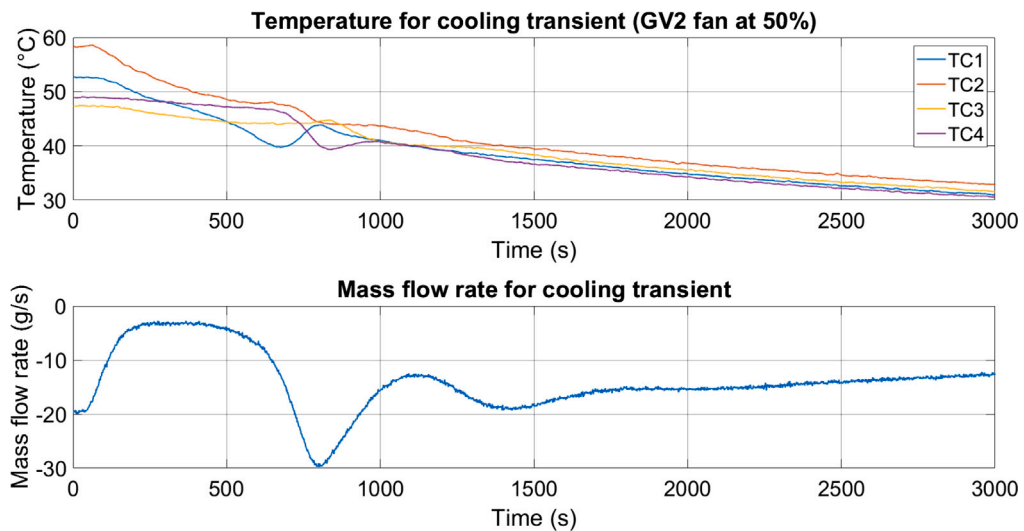


Fig. 13. GV2 cooling transient zoom.

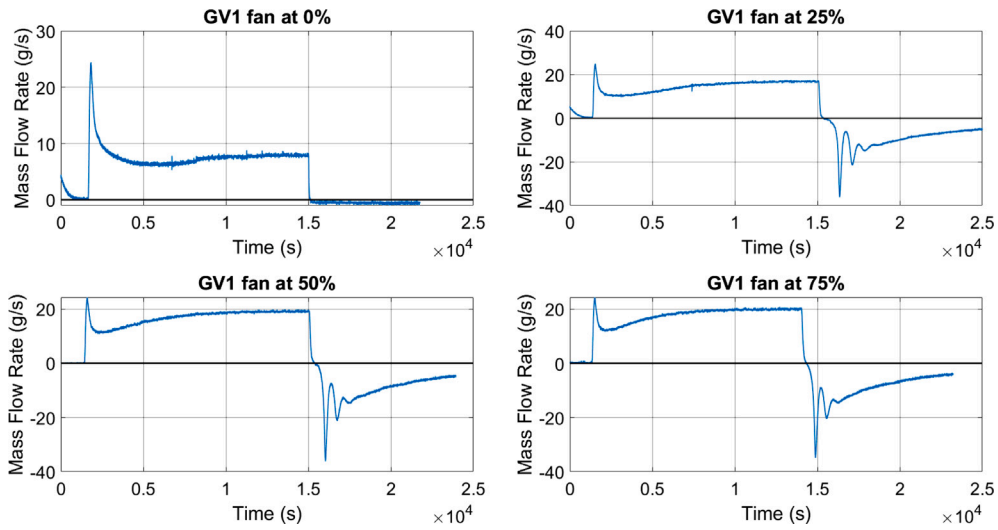


Fig. 14. GV1 results, mass flow rate evolution.

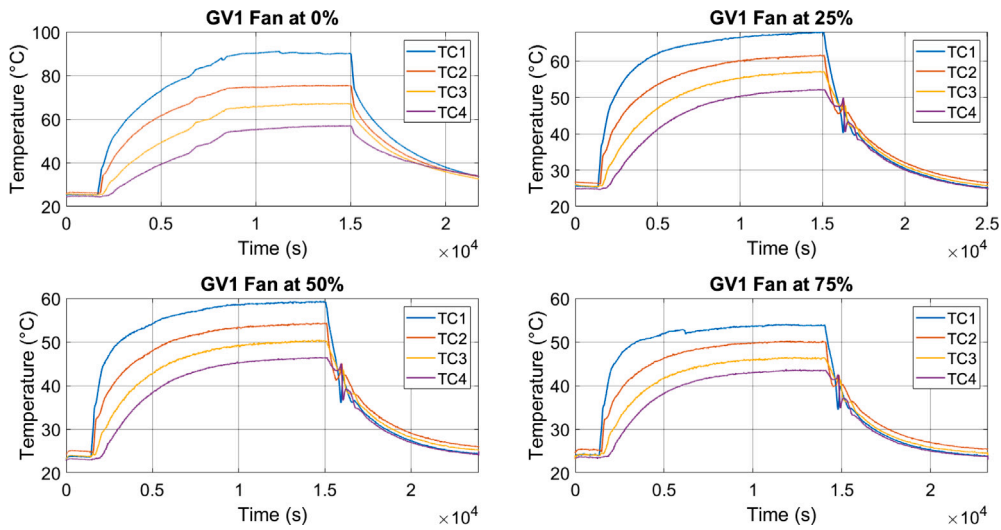


Fig. 15. GV1 results, fluid temperatures evolution.

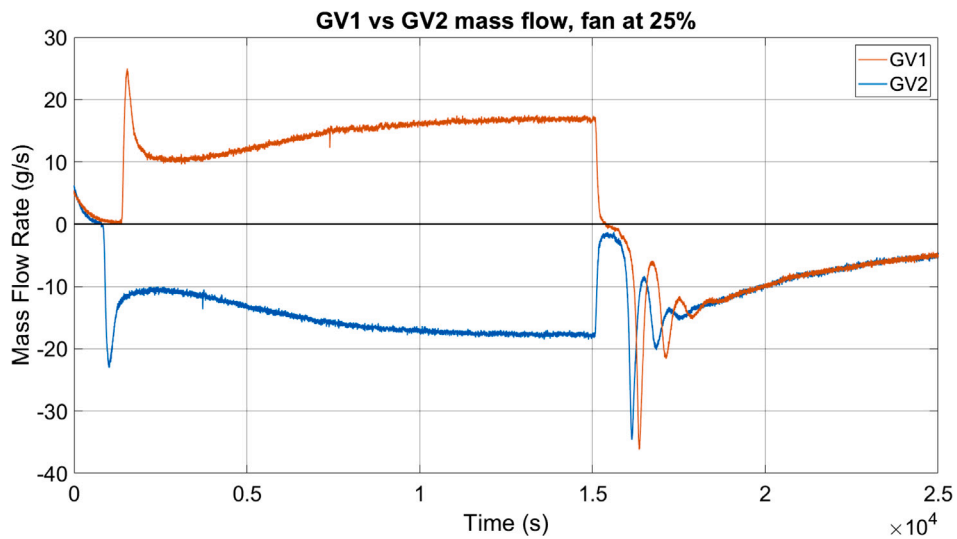


Fig. 16. Mass flow rate comparison — GV1 and GV2 cases at 25% fan speed.

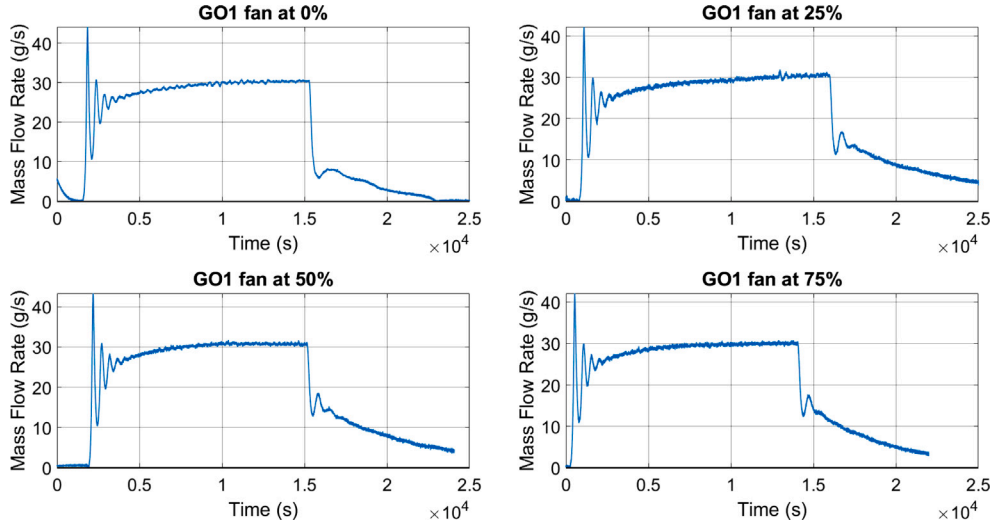


Fig. 17. GO1 results, mass flow rate evolution.

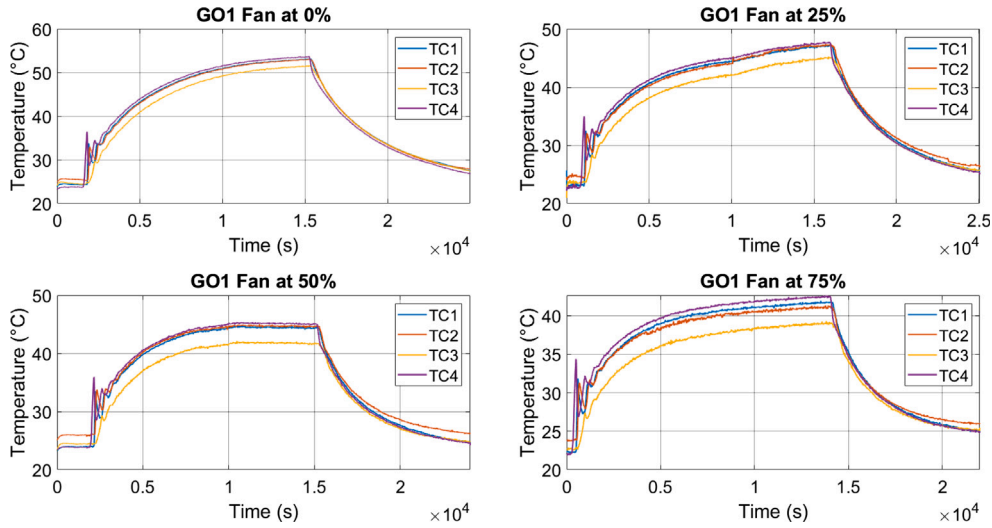


Fig. 18. GO1 results, fluid temperatures evolution.

transients, no sign of inversion is observed in the mass flow rate. Interestingly, TC1 and TC2 values are almost identical in all four HHHC cases, meaning that the cooler exchanges only a slight amount of heat. This is different compared to the two VHHC cases, where the cooler behaves as the primary heat sink of the facility. A plausible explanation for this peculiarity can be found in the small temperature differences of the heater (approximately 4 °C, confirmed by the low power that is absorbed by the fluid in this case) and between the fluid and the environment.

The HHHC cases also provide insights into the NCL dynamics, especially in the first part of the heating transient. In Fig. 19 the wave packet flow characteristic of NCL (Welandar, 1967) can be observed, where the hot and cold packets circulate in the system. In the experiment, TC4 is the outlet temperature of GO1 (the heater) and is the first one to see an increase in temperature. However, it decreases as soon as the hot fluid packet leaves the section headed for TC1, losing heat. The hot fluid packet then leaves TC1, passes through TC2 without losing heat, and then goes to TC3, losing the highest amount of heat. Thus, in the initial heating phase, GV2 acts as the cooling section of the experiment.

The temperature difference between TC2 and TC3 across the GV2 section is almost equal to the one between TC3 and TC4 across GO1.

It is also interesting to observe that the mass flow rate in these experiments always flows clockwise, despite the symmetric configuration. This can be explained again by the non-uniform distribution of fittings (and thus pressure losses) in the facility, which could lead to a preferred flow direction.

3.1.4. Experimental results: DH case

Last, the results of the DH case are presented. The setup is the same as in the VHHC and HHHC cases, but power is provided to all three heating legs at once. The experimental results are reported in Figs. 20 and 21 respectively for the mass flow rate and the temperatures. In these experiments, values of the cooling fan speed lower than 50% were not used since the fluid temperature reached the imposed safety limit of 95 °C.

During the heating transient, the mass flow rate curve exhibits a very pronounced oscillating behavior even after the initial transient, with peaks that resemble a delta-shaped function as high as 100 g/s. The oscillating nature of the transient is also confirmed by the temperature graph shown in Fig. 21. However, contrarily with the other cases, both the mass flow rate and the temperature curve of the DH case go smoothly to rest in the cooling transient. It is also interesting to observe the similarities with the HHHC case in the flow's direction, which is

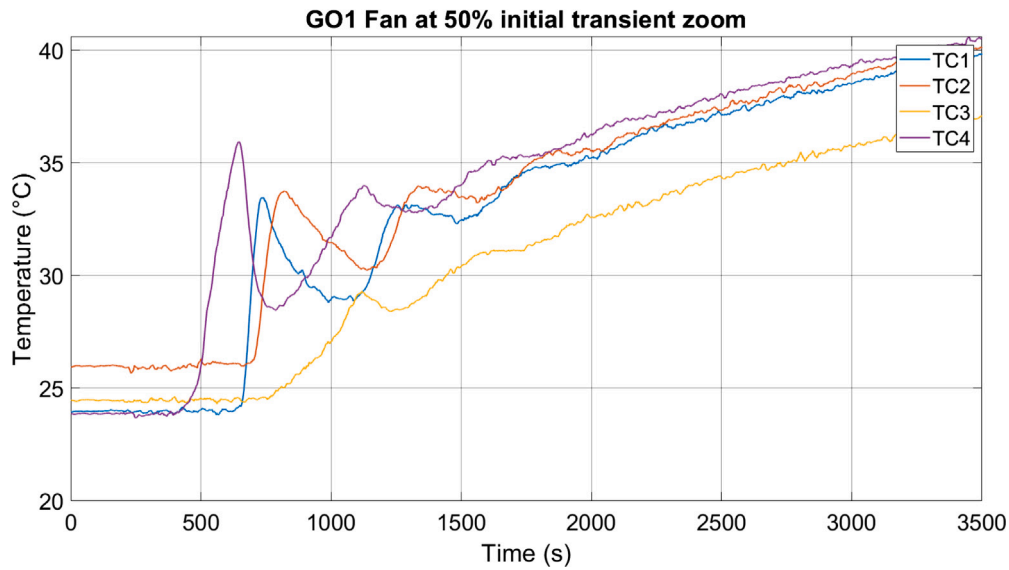


Fig. 19. Temperatures zoom during the initial transient of GO1 case.

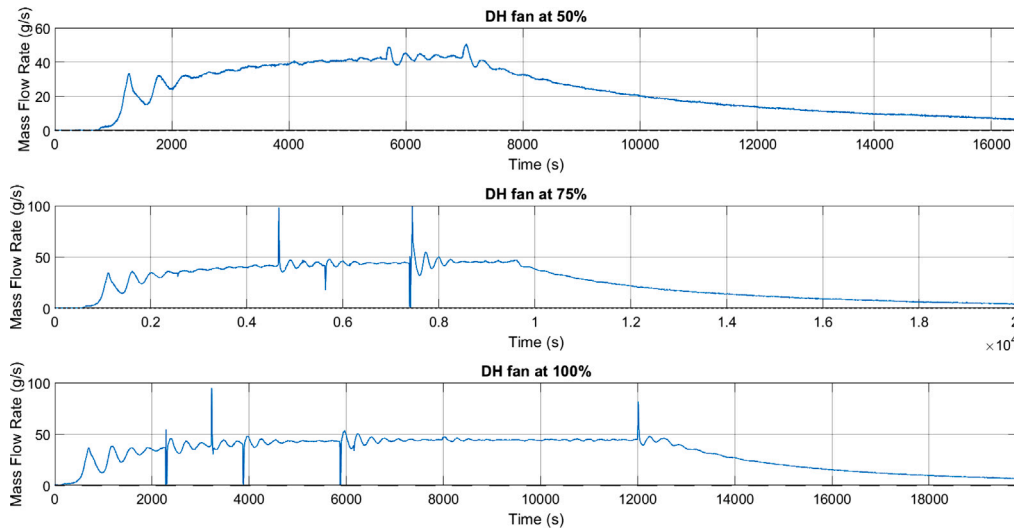


Fig. 20. DH results, mass flow rate evolution.

always clockwise. The preference, in this case, could also be because of having values of the heating power slightly different between the two vertical legs, leading to a preferred direction. The experiments show a very peculiar behavior of the facility: the horizontal leg GO1, despite providing heat, acts as a heat sink with a temperature difference similar to the one in the cooler. This peculiarity is shown in greater detail in Fig. 22. In particular, the temperature difference between TC1 and TC2 (cooler) is almost equal to the one between TC3 and TC4 (GO1). This can be explained by considering the absence of insulation and the relatively high power provided to the facility. The combined action of these two factors leads the temperature of the water to achieve higher values compared to the other cases (and hence increases the heat losses).

4. DYNASTY 1D model comparison with experimental results

In this section, the data found in the experimental campaign shown in the previous Section 3 are used to tune and validate, for the first time, the DYMOLA[®] DYNASTY models. In this comparison, the primary focus is the heating transient to characterize the capabilities of the

model to simulate the behavior of the experimental facility. However, since data is also available for the cooling, the comparison is extended to the whole experiment. In this comparison, the main FOMs (Figure Of Merit) are the mass flow rate and the temperature differences across the DYNASTY cooler.

The model used in this comparison is the DYNASTY final model reported in Fig. 4. In the first comparisons, the DYMOLA[®] model adopted the power calculated with energy balance at steady-state. Since the DH cases could not achieve steady-state during the experiments, there is no information about the delivered power, so they were excluded from the comparison in this paper. This model uses two different correlations to determine the value of the heat transfer coefficient in the cooler: for the case with the fan turned off the Churchill–Chu correlation is used, whereas when the fan is turned on the correlation used is the Churchill–Bernstein DeWitt and Lavine (2007). The calculated heat transfer coefficients of the heated sections have been tuned to achieve higher fidelity with the experiments. However, both the tuned and the non-tuned models are still not able to predict correctly the behavior of the cooling transient, except for the HHHC case. This is probably because of differences in geometry between the model and the facility

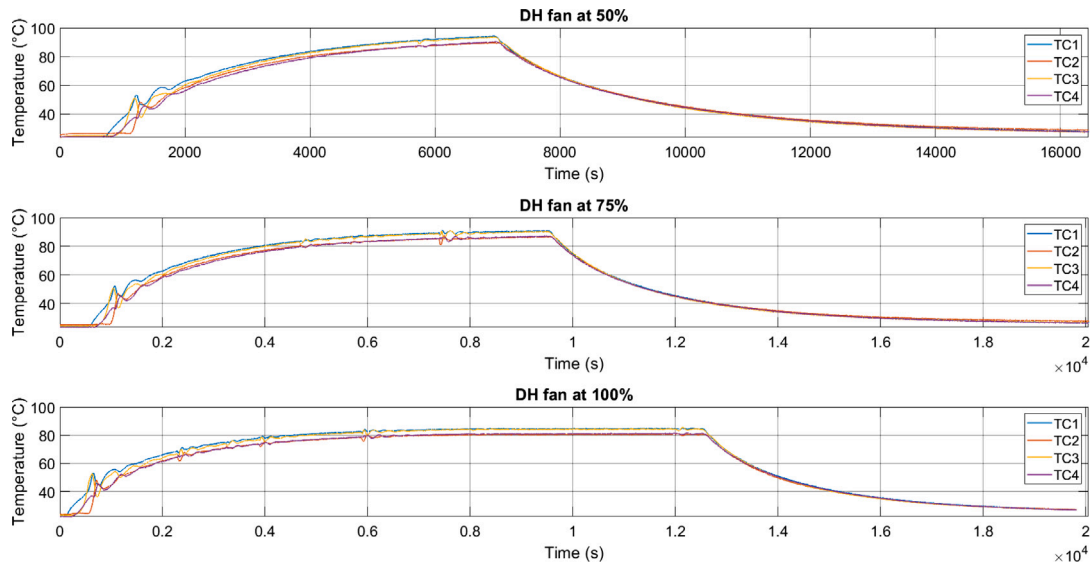


Fig. 21. DH results, fluid temperatures evolution.

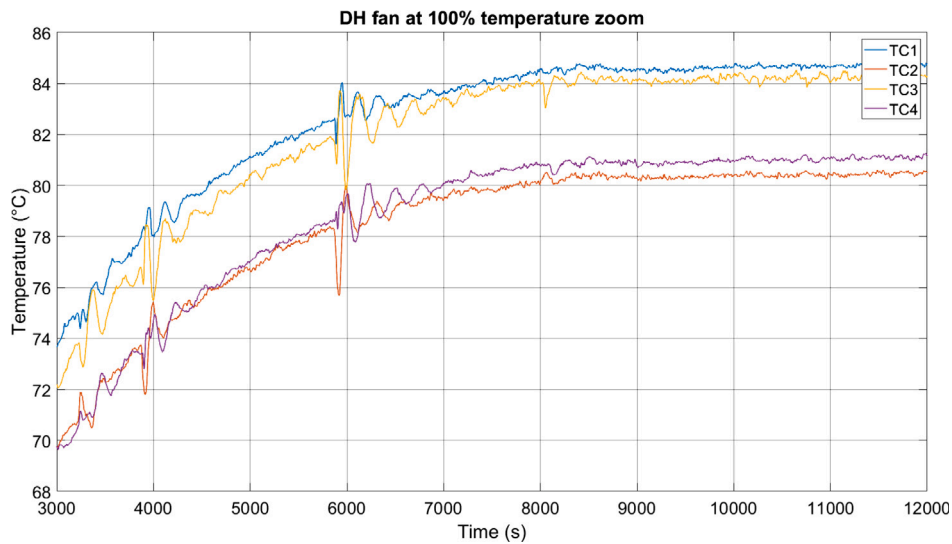


Fig. 22. DH Temperatures zoom.

that particularly affects the vertical configuration, to the adopted fluid heat transfer correlation (Cauzzi, 2019; Cammi et al., 2016b; Pini, 2017), and to the asymmetrical distribution of fittings.

To account for errors because of measuring, data acquisition, and electrical noise, error bands have been introduced in the graphs for the quantities shown, ± 2.15 °C for the temperature differences and ± 0.2138 g/s for the mass flow rate measured. These errors have been found by considering the tolerance of the sensors and by approximating “a posteriori” the errors of the data acquisition chain. The procedure for the mass flow rate is: the maximum error of the sensor (0.15% of the maximum mass flow rate value for each case) is used to calculate the standard deviation of each experimental case. Then the standard deviations are summed in quadrature to get the total sensor error of the mass flow rate. The estimation of error because of the data acquisition is performed by calculating the standard deviation between the experimental curve and a smooth one. The two errors (the sensor one and the estimated one) are then square averaged to achieve the total error used for the comparison (Moffat, 1988). The same approach was adopted for the thermocouples error estimation (Moffat, 1988). The absolute error of the sensors is already available and is equal to

± 1.5 °C (ELSI, 1998). The parameters and the provided power used for the different cases are:

- 1800 W is provided to both VHHC cases (GV1 and GV2);
- 825 W is provided for the HHHC cases;
- the air temperature used in the simulations is 24 °C;
- the convective heat transfer coefficient for all the legs has been multiplied by 2, whereas the radiating part of the heat losses has not been changed;
- the DYNASTY “naked” configuration is adopted, which means that both the model and the experiment are operating without the insulating layer;
- the DYNASTY model *PressDrop* *K* values have been increased, with the exception of *PressDrop* *4* that represents the MFRM, to better match the real facility valves, elbows, tees and flanges;
- the DYNASTY storage tanks are considered in the model only as additional pressure losses.

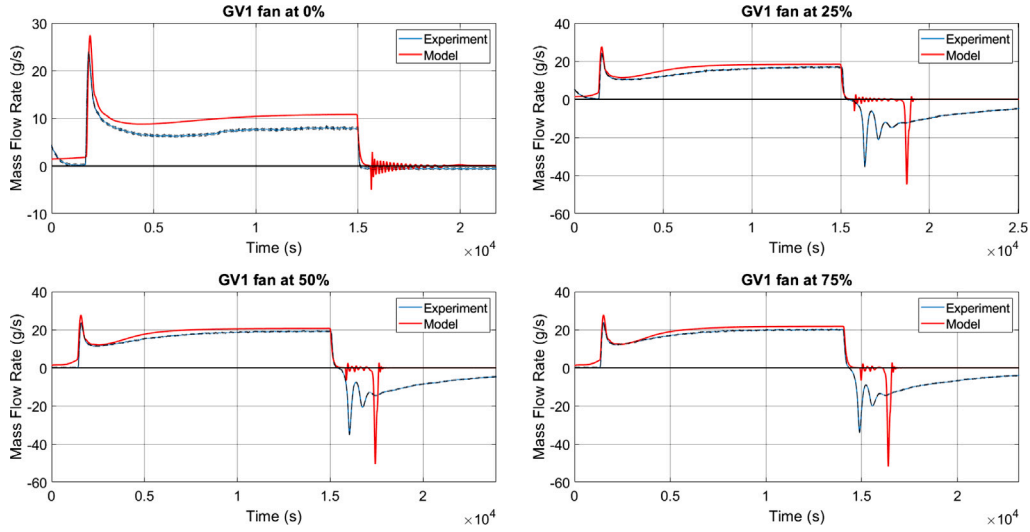


Fig. 23. GV1 mass flow rate evolution comparison.

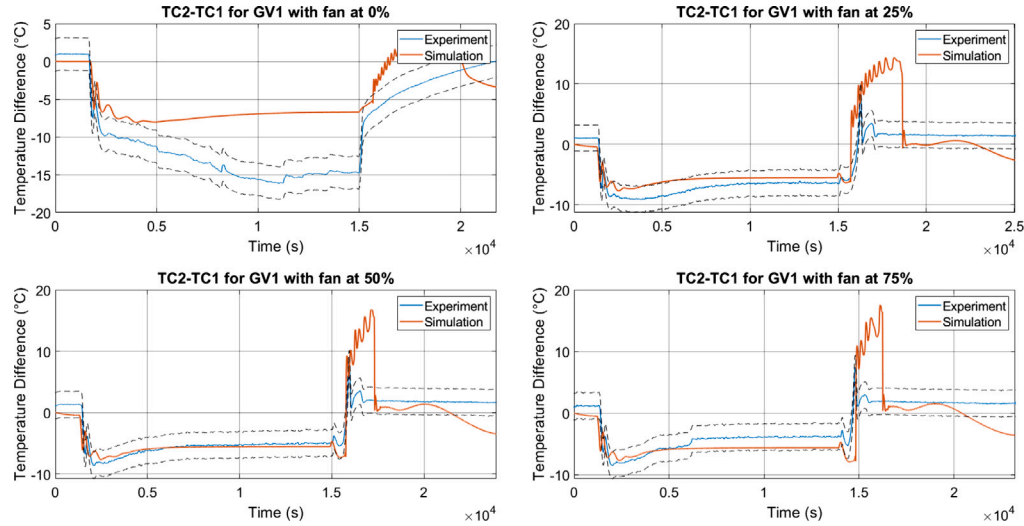


Fig. 24. GV1 Cooler Temperature evolution comparison.

4.1. DYNASTY model comparison results: VHHC case GV1

The first comparisons shown are the VHHC one with GV1 turned on reported in Figs. 23 and 24. The time transients are composed of $\sim 15,000$ s of heating and $\sim 10,000$ s of cooling. The following graphs collect the results of the comparison between the experimental data and the simulation results of the GV1 case, starting from 0% of fan speed and ending at 75%. Fan speed of 100% has not been used as experiments with this speed were not performed. These first comparisons showed good accordance between the model and the experiment, in both the mass flow rate and the temperature differences graphs. Indeed, the simulations mimic the curve behavior of the experimental cases for both the FOM. In the temperature difference graphs, the model results mostly stay inside the error boundaries, in particular during the heating transient. The only case where the model does not stay in the error boundaries in the heating transient is when the cooling fan is off. During the cooling transient, the model can predict the flow inversion that is characteristic of the GV1 experiments with the fan turned on. However, the cooling dynamics are still not properly simulated as the negative peaks lead to a sharp decrease in the mass flow rate, instead of the slow dampening observed in the experiments. In the case using fan speed of 0% the model exhibits a much more pronounced oscillatory

behavior around the experimental curve. The inability of the mode to predict the cooling transient behavior is probably, either because of the 3D effects (which occur during flow inversion as found in Battistini et al. (2021) that DYMOLA® (a 1D code) cannot model, or the presence of the open tank on top of the system, not present in the model. Additionally, the model geometry is not perfectly the same as the experimental facility concerning the position of the localized pressure drops (because of fittings and flanges). Finally, the correlation adopted for calculating the heat transfer coefficient between the pipe and fluid was developed for the heating transient, as previous work did not consider the cooling transient (Cauzzi, 2019; Cammi et al., 2016b). The contribution of all these issues then leads to the high discrepancies in the cooling transient between the experiment and the model.

4.2. DYNASTY model comparison results: VHHC case GV2

The second comparison shown is made with GV2, shown in Figs. 25 and 26. The time and power transients are set up as in the GV1 comparison. As in the previous case fan speed of 100% has not been used as experiments with this speed were not performed. The comparison results use increasing values of fan speed, starting from 0% and ending at 75% with an increasing step of 25%. Even in the GV2 cases, the 1D

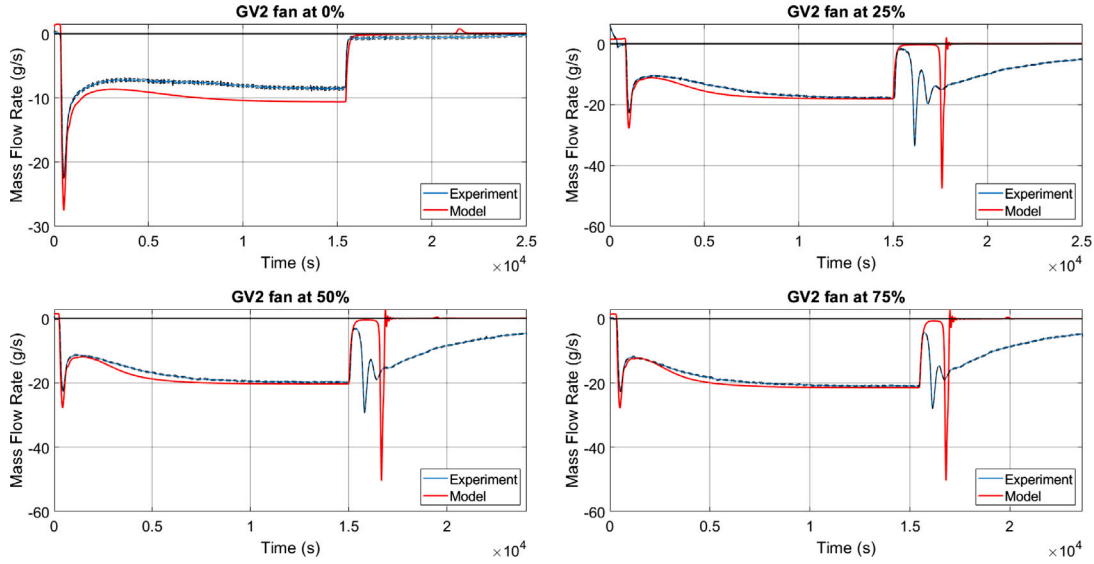


Fig. 25. GV2 mass flow rate evolution comparison.

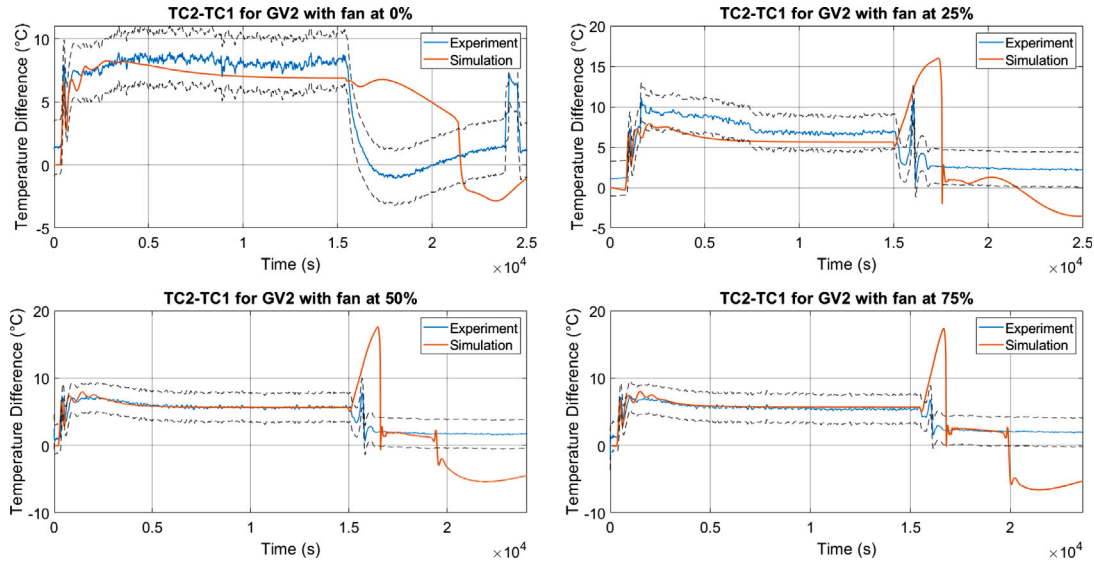


Fig. 26. GV2 Cooler Temperature evolution comparison.

model approximates the experimental curve. In particular, the model excels in the heating transient where, like in the symmetric cases GV1, it mimics the behavior and correctly estimates the value of both the mass flow rate and the temperature differences. However, in contrast to the GV1 case, the model simulates better the experiment during the cooling transient. In particular, the model shows the negative mass flow rate peak from fan speed of 25%, although it requires higher time than in the experimental case. As in the previous case, the poor performances in the cooling transient can be attributed to the tank, possible 3D phenomena, differences in geometry, and the heat transfer correlation. However, the model completes the simulations even if the MFR reaches almost zero during the cooling.

In the comparisons between the temperature differences, the model is showing good accordance with the experimental results, with the worst case being the one with the fan speed set at 0%. However, during the cooling transient, the simulated temperature differences diverge from the experimental ones, despite the not-so-different shape of the curves.

4.3. DYNASTY model comparison results: HHHC case GO1

The last comparison shown is the HHHC one with GO1 turned on. Figs. 27 and 28 collect the comparison results, with increasing fan speed from 0% to 75%. The heating power is applied as a step that starts at the same time as the experimental case at its maximum nominal value. The time transients used for the comparisons are set up as in the GV1 cases. As in the two previous comparisons fan speed of 100% has not been used as experiments with this speed were not performed.

In these last comparisons, the model is able to simulate correctly the behavior of both the mass flow rate and the temperature differences. In these cases, the curve shape is almost identical to the experimental one, although being shifted to higher or lower values depending on the fan speed. In the initial part of the heating transient, when the heaters have just been turned on, the simulation results exhibit a more pronounced oscillatory behavior than the experimental one for both the mass flow rate and the temperature difference. The model can simulate the behavior of cooling transients, except for the case with

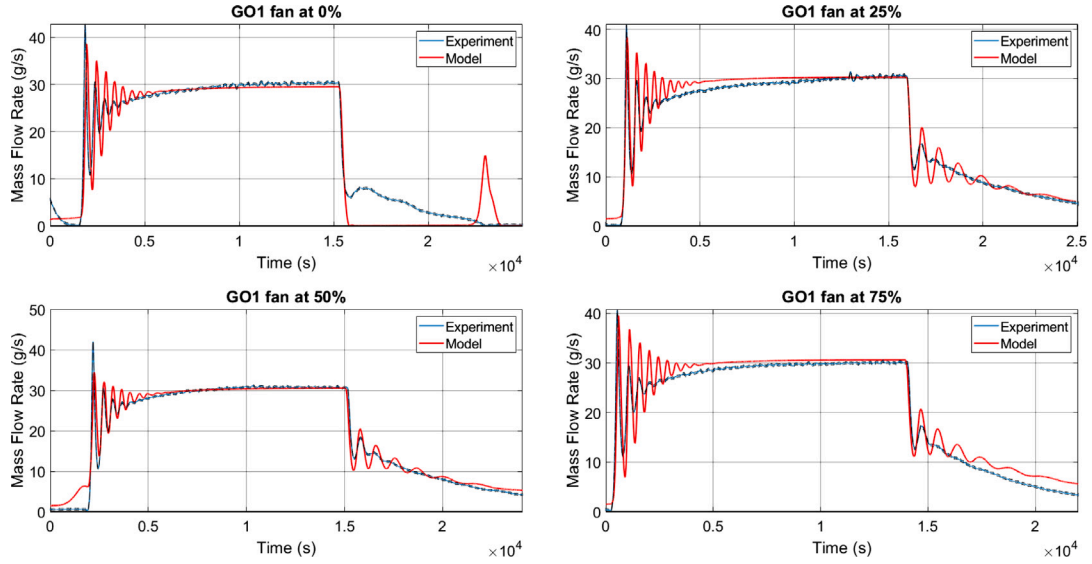


Fig. 27. GO1 mass flow rate evolution comparison.

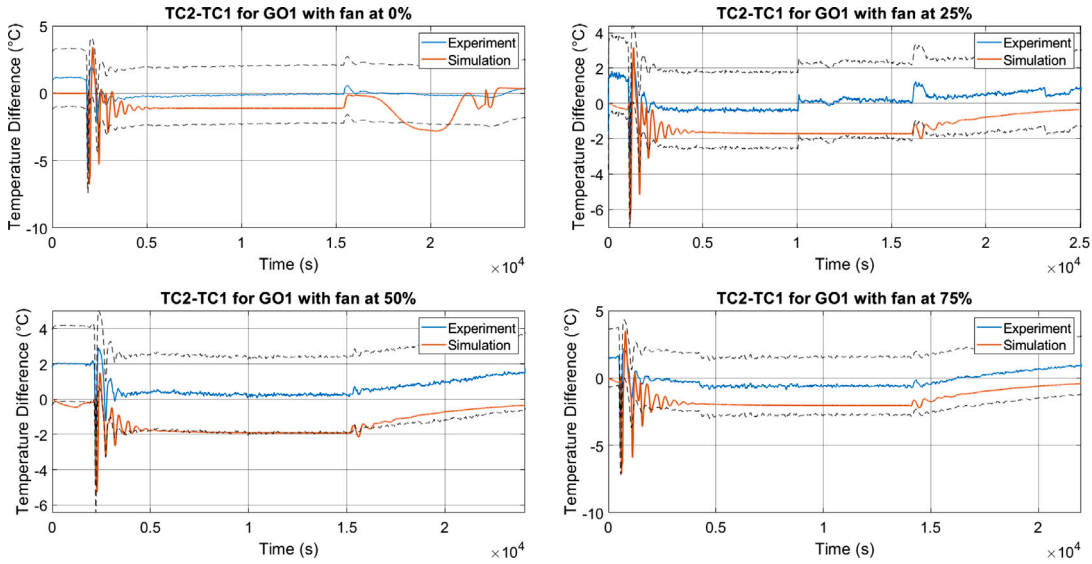


Fig. 28. GO1 Cooler Temperature evolution comparison.

the cooling fan off. However, despite the simulation curves staying inside the error band in most cases, the model mass flow rate curve presents more oscillations compared to the experimental one in both the heating and the cooling transients (especially in the initial part of the transients). In particular, in these comparisons the temperature differences predicted by the model exhibit promising results in both the heating and the cooling transients showing similar curve shape and value, although the simulation cannot predict the behavior of the cooling in the case with the fan turned off. In the case with a fan speed of 0%, the mass flow rate exhibit a peak after 20,000s during the cooling transient. This was found to be because of an increase in the temperature difference when the mass flow rate reached the minimum, which caused a build-up of potential energy that lead to a sudden restart of the fluid motion. As in the previous cases, the discrepancies between the model and the experiments could be found in the inability of DYMOLA® to model 3D phenomena (observed in Battistini et al., 2021), the presence of the DYNASTY tank, differences in geometry, and the fluid heat transfer correlation. The oscillation frequency in the mass flow rate curve in the cooling transient for both the model and the experiment are comparable, but the experiment dampens the

oscillations faster than the model. Also, in the cases with the fan at 50% and 75%, the model mass flow rate drifts from the experimental one despite similarities in the temperature difference graph.

4.4. DYNASTY model comparison summary

Finally, Tables 4 and 5 collects a summary of the outcomes of the comparison with a measure of the relative error in the mass flow rate graphs between the experiment and the model, differentiating between the heating and cooling transient. The formula used for the computation of the mean relative errors for Tables 4 and 5 is shown in Eq. (1). In this equation, the squared difference between the experimental and the model value for each second is divided by the square of the experimental value, getting a squared relative error. This error is then divided by the number of experimental values, which corresponds to the number of seconds of the experiments. The square root of this square averaged relative error leads to the value of the error that is reported in the tables following an approach similar to the one proposed in Moffat (1988). The errors for the heating transient have been computed neglecting the initial oscillations, which, especially in the HHHC case, lead to values

Table 4

Mean relative error for the heating transient in the different comparison cases with increasing values of fan speed.

Fan speed	GO1 heat	GV1 heat	GV2 heat
0%	0.0192	0.4000	0.2815
25%	0.0319	0.1450	0.0707
50%	0.0109	0.1150	0.0485
75%	0.0261	0.0959	0.0409

Table 5

Mean relative error for the cooling transient in the different comparison cases with increasing values of fan speed.

Fan speed	GO1 cool	GV1 cool	GV2 cool
0	N.A.	1.4839	0.7650
25%	0.1131	0.9567	0.9212
50%	0.1322	0.9562	0.9215
75%	0.3598	1.1791	0.8880

in the absolute error close to infinity; the same is true for the cooling transient, where, as soon as the experimental value reaches values near zero, the relative error value increase. In the tables, the relative error of the GO1 case with fan of 0% was neglected as values higher than 150% were found, as expected from the MFR comparisons in the cooling transient (Fig. 27).

$$err = \left(\frac{\sum \left(\frac{(\text{experimental value} - \text{model value})^2}{\text{experimental value}^2} \right)}{\text{number of experimental values}} \right)^{0.5} \quad (1)$$

5. Conclusions

In this work, the 1D modeling of the DYNASTY facility has been improved and preliminarily validated with the results coming from the first DYNASTY experimental campaign (Benzoni et al., 2022). The improved model was then thoroughly analyzed by varying the fan speed, the mass flow rate meter pressure loss coefficient k , and the external air temperature. These tests explored the behavior of DYNASTY under different conditions, and they represent the starting point for the subsequent realization of the experimental campaign.

The experimental campaign analyzes the facility behavior with different fan speed values for all the possible heating configurations (vertical and horizontal localized and distributed heating) at maximum power and without the insulation layer to characterize DYNASTY thermal losses (Benzoni et al., 2022). The results of the experimental campaign showed peculiar behavior in the DYNASTY facility. In the heating transient, the two VHHC cases behave as almost perfectly symmetrical cases showing similar values of the mass flow rate with the opposite sign. However, the major difference was observed in the temperature graphs where, for the GV1 case with a fan speed of 0%, the fluid reaches temperatures in the central part of the pipe of 95 °C. The GV2 case instead exhibited temperatures much lower despite delivering the same heating power. In the cooling transient, the facility behavior is almost the same for the two VHHC cases, which could mean that the heating part sets only the initial condition for the evolution of the system during cooling. In the GO1 case, the cooler presents a temperature difference between the inlet and outlet of less than 1 °C while the heater temperature difference is ~4 °C. However, the highest temperature difference is observed in the two vertical legs, meaning that the two naked legs behave as the primary heat sink in the GO1 transient. The cooling transient of the GO1 case exhibits a dampened oscillating behavior with a slow decrease in time, in contrast with the fast decrease of both the GV1 and GV2 cases. Finally, during the DH cases, the water temperature reached values as high as 95 °C with only

the one at 100% fan speed reaching a stationary state at ~85 °C. Also, in the DH experiments, the horizontal leg GO1 dissipates heat with similar performances to the cooler.

The comparison of the improved 1D modeling with the results of the experimental campaign shows satisfactory results for the heating transient for the different heating configurations as during the heating transient, the 1D model could predict in a good way the experimental results. This is remarkable for the models as Modelica (and thus DYMOLA®) is a simulation code not capable of simulating 3D phenomena. In particular, the model can catch the correct mass flow rate sign for all the configurations tested. The model can also simulate the facility behavior in the heating transient, both in terms of the temperature difference and mass flow rate. The simulations also provide an accurate prediction of both the temperature and the mass flow rate values, while showing improving margins in dampening the initial oscillations of the GO1 cases. However, when considering the cooling transient, there are significant mismatches between the experiments and the simulations. This mismatch is especially clear in the VHHC temperature difference graphs, where the model completely drifts from the experiment results. Despite this, the model still catches some dynamics of the VHHC cooling, as the mass flow rate curves show a high negative peak with a decrease to zero, even if shifted in time. The HHHC cases show promising agreement between the experiment and the model also during the cooling transient, having both a similar decreasing behavior and a matching temperature differences graph, even if shifted. Here, the most interesting graphs are the mass flow rate one, where, except for the case with the fan at 0%, the curve exhibit an oscillating behavior around the experimental one. Still, increasing the fan speed, the model drifts more and more from the experiment.

Results of this preliminary validation set the path for future developments that will include further experimental analysis of the cooling phase, along with improvements in the simulation capabilities for both the cooling transient and the DH configuration. The validation process will continue on the insulated facility with different values of heating power. The facility will also be tested with a different working fluid. In particular, a mixture of water and glycol will be used to mimic the thermal hydraulics proprieties of the molten salt (i.e., high Prandtl number fluids). Future works will also regard the experimental facility itself: as previously said, the insulator layer will be reinstalled; more measuring systems will be added, in particular the pressure meter and flow rate meter.

Data availability

Data will be made available on request.

Appendix. DYNASTY modeling

A.1. DYNASTY modelica models description

This appendix describes the development process that led to the final DYNASTY models used in this work. The first 1D Modelica models used to simulate DYNASTY were developed during previous works and are shown in Fig. A.29 (Cauzzi, 2019; Pini, 2017). This model uses modified components from the standard ThermoPower library (Casella and Leva, 2003, 2005), storing the changed components in the custom-made ThermoPowerIHG library. The different components in the model shown in Fig. A.29 are:

- *L2, L7, L4* are the models of the DYNASTY pipes and account for heating in the facility by imposing the heating power on the outer wall of the pipe;
- *Cooler* is the model of the DYNASTY cooling section: this model fixes the temperature at the outer wall of the pipe;

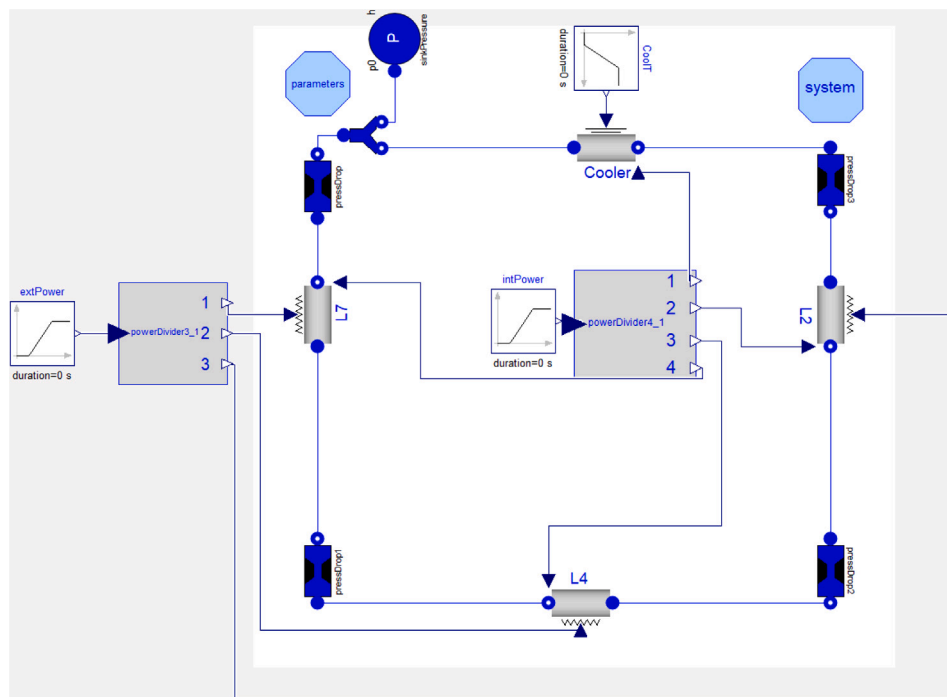


Fig. A.29. DYNASTY simplified water model.

- *PressDrop* 1 to 4 represent the DYNASTY elbows localized pressure losses;
- *PowerDivider* is a component that takes an input signal and divides it according to user-imposed weights to obtain multiple output signals, allowing for the HHHC, VHHC, and DH heating configurations;
- *SinkPressure* sets the pressure on top of the loop;
- *CoolT* is a source block that sets the temperature of the outer shell of the cooler walls;
- *extPower* is a source block that gives the signal of the EH source
- *intPower* is the source block that provides the HG to the system;
- the two blocks *parameters* and *system* set the DYNASTY thermo-physical properties and the initial conditions of the simulation.

The DYNASTY pipe model is shown in Fig. A.30 with the different components that build the model. This pipe model is quite similar to the actual DYNASTY pipe; however, it suffers from a drawback in the heat losses model between the pipe and the environment. The different components of this pipe model are then the following:

- *customFlow1DFEM* is the fluid dynamics model of the fluid inside the pipe, which solves the balance equations (Cauzzi, 2019). This component also accounts for the elevation of the outlet compared to the inlet;
- *metalTubeFEM* is the model of the pipe walls' thermal capacity and resistance, which represents the heat exchange between the fluid inside the pipe and an external heat source;
- *heatSource1DFEM* represents the externally imposed heat flux on the external surface of the pipe;
- *iHG1DFEM* represents the internal heat generation;
- *EHF* is a real input port that provides the signal for the imposed heat flux;
- *IHG* is a real input port that provides the signal which imposes the value for the internal heat generation;
- *Flange A* and *Flange B* are the flow connector ports which connect the different parts of DYNASTY and account for the motion of the fluid.

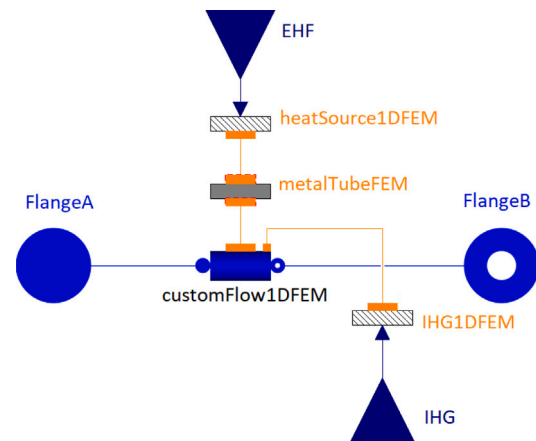


Fig. A.30. DYNASTY pipe model.

The DYNASTY models are then made by connecting the different required components. For example, Fig. A.31 shows the flow connection between two different pipe components, the *Cooler* and *L8*. When a connection between compatible pipe models is present, Modelica automatically implements mass, energy, and momentum balance equations between the two ends of the pipes. This connection allows the fluid to flow between the two components, carrying all the information to determine its thermodynamic state.

These DYNASTY Modelica models ignore several aspects of the facility. In particular, these models do not account for the concentrated pressure losses because of the Mass Flow Rate Meter (MFRM), the heat losses between the pipes and the environment, the fan in the cooler, and they can strictly model only single-phase fluids. Thus, one of the first improvements in modeling the DYNASTY facility was the implementation of the pressure losses because of the PROMASS F80 DN25 MFRM (Endress-Hauser, 2017), not accounted for in the models available in the literature. Preliminary calculations estimated these

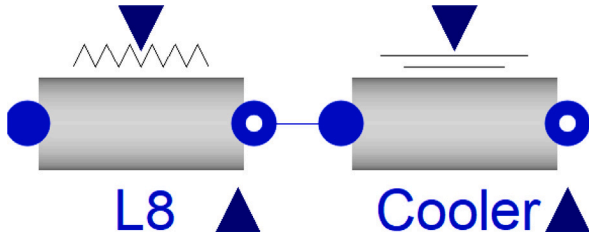


Fig. A.31. Pipes connection example.

Table A.6

MFRM parameters.

Reynolds number (Re)	$Re = \frac{2 \cdot \dot{m}}{\pi \cdot d \cdot \nu \cdot \rho}$
$Re \geq 2300$	$\Delta p = K \cdot \nu^{0.25} \cdot \dot{m}^{1.85} \cdot \rho^{-0.86}$
$Re < 2300$	$\Delta p = K1 \cdot \nu \cdot \dot{m} + \frac{K2 \cdot \nu^{0.25} \cdot \dot{m}^2}{\rho}$

Table A.7

MFRM parameters values.

DN	$d(m)$	K	$K1$	$K2$
25	$12.00 \cdot 10^{-3}$	$1.90 \cdot 10^6$	$6.40 \cdot 10^6$	$4.50 \cdot 10^5$

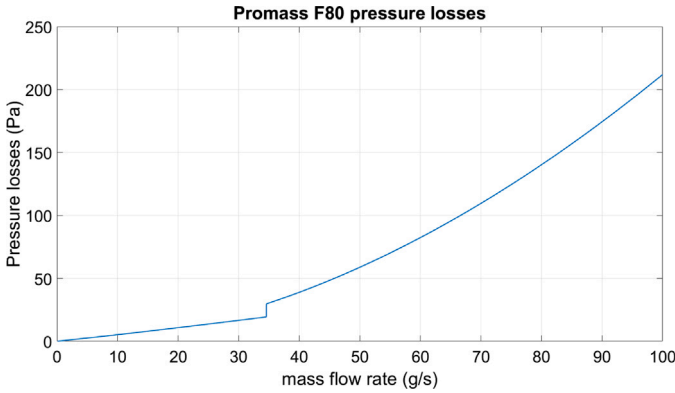


Fig. A.32. MFRM pressure losses.

losses to ~ 3 times the entire loop ones. Thus, the model was changed to account for the MFRM pressure losses as follows:

- the $L4$ component was split into two components $L4$ and $L5$ to better represent the DYNASTY design;
- $sensW$ is a mass flow rate sensor taken from the ThermoPower library;
- $PressDrop4$ represents the localized pressure drop of the MFRM;

To implement the MFRM, the correlation provided by ENDRESS+HAUSER (Endress-Hauser, 2017) and reported in Table A.6 was used; Table A.7 reports the reference values. K , $K1$, and $K2$ are constant, which depends on the type of MFRM considered, $\dot{m}$ is the mass flow rate, d is the inside diameter of the MFRM measuring tubes, ν is the fluid kinematic viscosity, and Δp are the pressure losses of the MFRM. After the pressure loss curve of the MFRM was plotted to the operative DYNASTY mass flow rate range (Fig. A.32), from 0.1 g s^{-1} to 100 g s^{-1} , interpolation was performed to obtain a coefficient suitable for the use in the DYMOLA[®] model. The discontinuity in the graph is because of the change in the correlation used to calculate the pressure losses (Endress-Hauser, 2017).

Other improvements in the model are the implementation of the heat losses modeling in the DYNASTY pipe components. The improved model of the DYNASTY pipe is shown in Fig. A.33 where the components T_{air} , $doubleHeatSource1DFEM$, $metalTubeFEM_insulator$, and

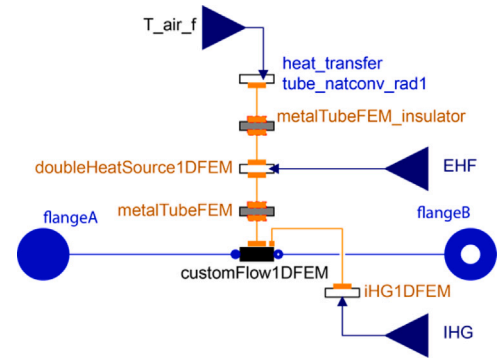


Fig. A.33. Modelica view of the DYNASTY pipe with heat losses.

$heat1_transfer_tube_natconv_rad1$ allow for including the thermal losses in the pipe model. Their function is listed below:

- T_{air} functions as an external input that gives the signal for the air temperature: if the flag “use T_{air} input” in the pipe model is on it sets the values of T_{air} ;
- $metalTubeFEM_insulator$ models the presence of the insulation layer on the pipes acting as additional thermal inertia between the facility and the environment;
- the $doubleHeatSource1DFEM$ component models the heat exchange between the insulated inner surface, the pipe outer surface, and the external heat source as shown in Eq. (A.1). In the equation, $\Phi_{pipe,ii}$ is the heat flux to the pipe ii -th node, $\Phi_{ins,ii}$ is the heat flux to the insulator ii -th node, Q is the provided power, D is the interface diameter, and L is the interface axial length. The temperature is continuous at the interface hence the pipe interface temperature $T_{pipe,ii}$ and the insulator interface temperature $T_{ins,ii}$ are equal;
- $heat1_transfer_tube_natconv_rad1$ models the heat exchange between the outer insulated walls and air. As shown in Eq. (A.2), the heat convective coefficient uses the Churchill–Bernstein correlation, while for the radiative term, the small surface-large isothermal enclosure approximation is used (Bergman et al., 2011). In Eq. (A.2), Φ_{ii} is the heat flux to the external pipe surface at the ii -th node, ϵ is the insulator emissivity, σ_{SB} is the Boltzmann constant, h_{conv} is the convective heat transfer coefficient, $T_{w,ii}$ is the external pipe wall temperature at the ii -th node, and T_{air} is the air bulk temperature.

$$\begin{cases} \Phi_{pipe,ii} + \Phi_{ins,ii} = -\frac{Q}{D \cdot L \cdot \pi} \\ T_{pipe,ii} = T_{ins,ii} \end{cases} \quad (A.1)$$

$$\Phi_{ii} = \epsilon \cdot \sigma_{SB} \cdot (T_{w,ii}^4 - T_{air}^4) + h_{conv} \cdot (T_{w,ii} - T_{air}) \quad (A.2)$$

To further improve the models of the experimental facility, the cooler component was improved as well. The previous cooler model adopted an imposed wall temperature, coherently with the stability map analysis (Ruiz et al., 2015). However, the approximation of the imposed cooler temperature proved quite limiting, especially for comparing the simulation results with the experimental one. Thus, the $Heat_transfer_coefficient_Source$ component was developed. This component accounts for the convective heat exchange between the cooler and the environment. Thus, it becomes necessary to include two new input values V_{air} and T_{air} for computing the dissipated heat from the cooler. The model then uses the Churchill–Bernstein correlation (Bergman et al., 2011) to calculate the average Nu number and the average convective heat exchange coefficient $\bar{h}$ between the cooler

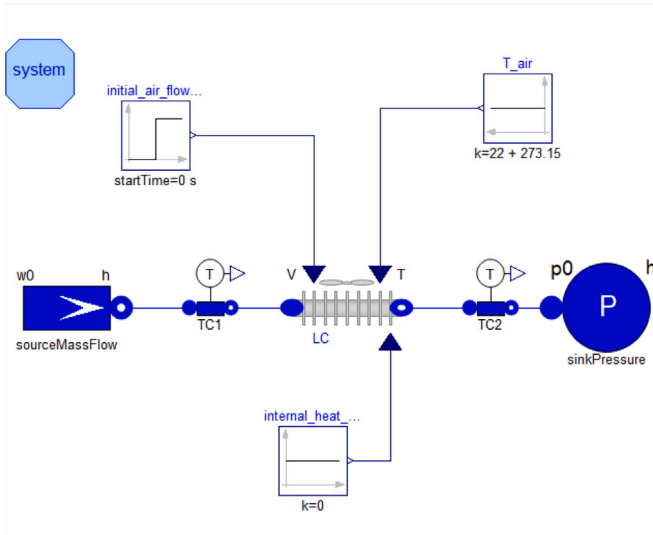


Fig. A.34. Cooler Test setup.

and the air, as in Eq. (A.3).

$$\bar{h} = \frac{\overline{Nu} \cdot k}{D}$$

$$\overline{Nu} = 0.3 + \frac{0.62 Re_D^{1/2} Pr^{1/3}}{[1 + (\frac{0.4}{Pr})^{2/3}]^{1/4}} [1 + (\frac{Re_D}{282000})^{5/8}]^{4/5} \quad (A.3)$$

recommended for $Re_D \cdot Pr \geq 0.2$

In order to account for the increase in the exchange surface because of the fins in the cooler, the procedure given by Bergman et al. (2011) and reported in Eq. (A.4) and Eq. (A.5) is used, where N is the total number of fins, A_{fin} is the single fin area, $A_t = A_b + N A_{fin}$ is the total area, A_b is the area of the pipe without fins and η_{fin} is the fin efficiency that is calculated using Eq. (A.5):

$$Q = h \cdot A_t \cdot [1 - \frac{N A_{fin}}{A_t} (1 - \eta_{fin})] \Delta T \quad (A.4)$$

$$\eta_{fin} = C_2 \frac{K_1(mr_1)I_1(mr_{2c}) - I_1(mr_1)K_1(mr_{2c})}{I_0(mr_1)K_1(mr_{2c}) + K_0(mr_1)I_1(mr_{2c})} \quad (A.5)$$

$$C_2 = \frac{(2 \frac{r_1}{m})}{(r_{2c}^2 - r_1^2)}$$

where K_i and I_i are i -order modified Bessel functions of the first and second kind respectively, described in Appendix B of Bergman et al. (2011), r_1 is the pipe outer radius, $r_{2c} = r_2 + (\frac{t}{2})$ is the corrected radius for a convective tip, r_2 is the fin radius and m is a coefficient that depends on the fin heat exchange.

To compare the performances of the finned and non-finned cooler, a simple test model was developed using water as medium. Fig. A.34 shows the test model; the components used to build it are the following:

- $TC1$ and $TC2$ are two thermocouples taken from the ThermoPower library;
- LC is the cooler component;
- $sourceMassFlow$ is a source component for the water mass flow rate, which also sets the inlet temperature of the fluid;
- $sinkPressure$ in this model acts as a sink for the mass flow rate;
- $initial_air_flow$ and T_air are two inputs, which set the working conditions of the cooler in terms of airflow rate and air temperature.

The test then involved delivering the same water mass flow rate of $30 \frac{m^3}{s}$ at $60^\circ C$ at time zero to both the finned and non-finned cooler

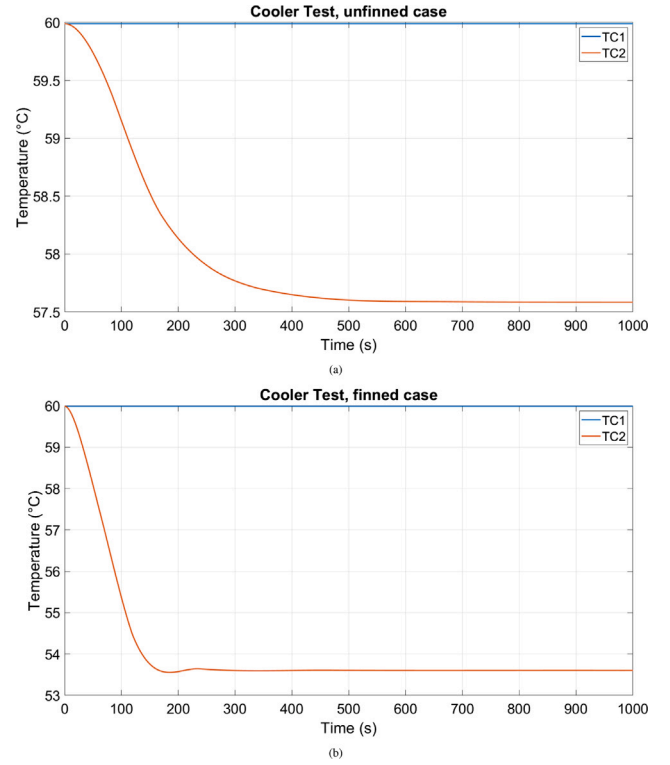


Fig. A.35. Cooler test, (a) non-finned case and (b) finned case.

models. In the simulation, the temperature of $60^\circ C$ was also imposed on the complete model to initialize the simulations. The airflow rate of the cooler is set at $4 \frac{m^3}{s}$ while the air temperature is equal to $22^\circ C$. The simulations are then run for 1000 s, collecting the two thermocouples signals $TC1$ and $TC2$. Fig. A.35 shows the temperature results, which show the impact of the finned pipe versus a non-finned one: the higher heat transfer coefficient of the finned cooler leads to a lower steady-state value of outlet temperature $TC2$ with faster time evolution. The small oscillation in Fig. A.35(b) is because of an initial overshoot of the model in the exchanged heat, as these simulations consist in assigning a fixed input value to the cooler and then observing the steady state value found in the free evolution of the model.

A.2. DYNASTY numerical algorithm performance test

The improvements made in the DYNASTY Modelica modeling to achieve a higher fidelity when simulating natural circulation phenomena caused an increase in the computation burden of the simulation. Since the simulation of natural circulation with possible flow inversion, the time required to complete the simulations with DYMOLA® default algorithm reached unsustainable levels, and this required the characterization of the performances of some of the stiff numerical integration algorithms present in DYMOLA®.

These tests analyze the algorithm performances by plotting the mass flow rate and the CPU time. The CPU represents the time taken by the CPU at each time step to simulate the model, taken as a figure of merit for the test. CPU time as FOM allows for classifying the time performances of the different algorithms in real-time simulations for control of the experimental facility. The performance tests were performed using an AMD® RYZEN® 5 2600 Central Processing Unit (CPU) having six cores and 12 threads with a maximum clock speed of 3.75 GHz. It is worth noting that DYMOLA® can run multi-core simulations by setting the flag “Advanced.ParallelizeCode=true”. This option, on average, reduces the computational time by half despite the

Table A.8
Numerical integration algorithms test summary.

ALGORITHMS	DASSL	RADAU2A	SDIRK34HW	ESDIRK45A	LSODAR
Method	Linear-multi-step	Single step	Single step	Single step	Linear-multi-step
Stiff	Stiff	Stiff	Stiff	Stiff	Both
implicit/explicit	Implicit	Implicit	Implicit	Implicit	Both
Order	1–5	5	4	5	1–5
Complete Simulation	No	Yes	Yes	Yes	No

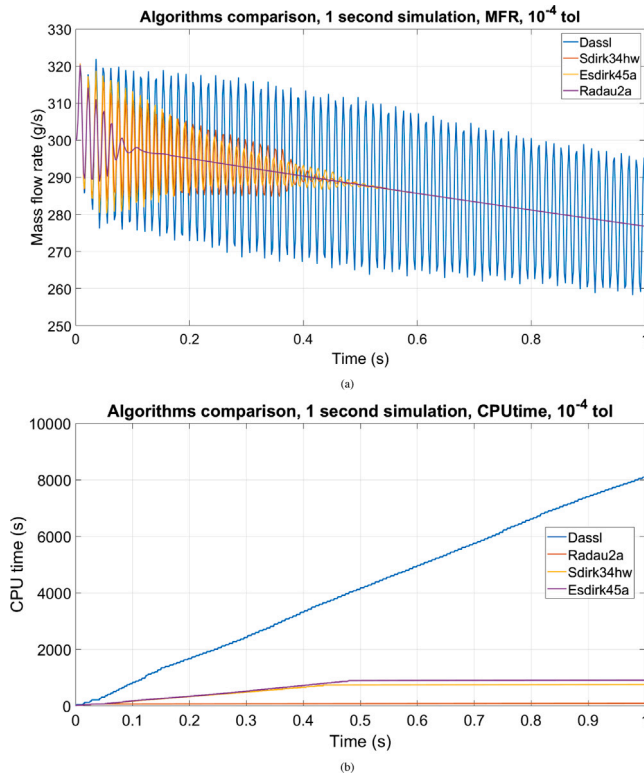


Fig. A.36. Mass flow rate A.36(a) and CPU A.36(b) time for one second simulation, 10^{-4} tol, DH.

increased time required to translate the model. The methods tested are the following:

- DASSL (Petzold, 1982), a multi-step implicit numerical integration algorithm based on the Backward differential formula (BDF);
- LSODAR (Hindmarsh and Petzold, 2005; Cellier and Kofman, 2006) a multi-step explicit/implicit numerical integration algorithm based on the Adam method for non-stiff problems and automatically switch to BDF methods when required;
- Explicit Singly Diagonal Implicit Runge–Kutta (ESDIRK)45as (Jørgensen et al., 2018), Radau2a (Cellier and Kofman, 2006), and Singly Diagonal Implicit Runge–Kutta (SDIRK)34hw (Jørgensen et al., 2018) are all single step implicit Runge–Kutta (RK) methods.

In particular, the tests highlighted that only SDIRK45a, Radau2a, and SDIRK34hw completed the simulation in a reasonable time. The LSODAR and the DASSL algorithms could not finish the simulations. Because of the complexity of the DYNASTY models and because the DAE system is translated into a stiff ODE system to be solved (Cellier and Kofman, 2006; Pantelides, 1988), no tests were performed with explicit methods (see Table A.8).

The results of the tests, in particular the ones in Fig. A.36, show the first second of the simulation for a DH case, with a user-defined

tolerance of 10^{-4} . While the single-step methods mitigate the initial oscillations very well, with Radau2a outclassing the other RK methods, DASSL instead cannot dampen the oscillations, requiring an enormous amount of time (8000 s) to simulate just one second of real-time. The very high computation time necessary is compatible with chattering in the simulations. Chattering is the degradation of the simulation performance when many events occur, which forces the algorithm to use a shorter step size than necessary, leading to a drastic increase in the computation time (Tiller, 2019). Chattering is a systematic issue that should affect the other numerical integration algorithm, not only DASSL. However, The DASSL algorithm is a linear multi-step method that struggles when step size needs frequent change (Cellier and Kofman, 2006). When simulating DYNASTY, many step size adjustments are required at the start of the simulation to solve the initialization problem. Another problem of DASSL is that when the tolerance is too low, the algorithm does not overcome the initial perturbation and keeps oscillating, as in chattering, requiring simulation time in the order of days to simulate a couple of hundreds of seconds. The simulations also gave results not compatible with the actual DYNASTY behavior. This finding agrees with the literature, Liu et al. (2010), Cellier and Kofman (2006), where coarser tolerances must be defined to use DASSL, which leads to decreases in simulation accuracy. The outcome of these tests is that the Radau2a integration algorithm was chosen as the default algorithm for simulating the current DYNASTY models.

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